



Sexual Behavior Repertoire in Sub-Saharan Africa and Risk of Human Papillomavirus Infections and Associated Cancers

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Outline

- A brief description of sub-Saharan Africa
- Sexual behavior repertoire
 - definition and coverage
 - Concept of sexual risk behaviors
- Oral and anal sex in sub-Saharan Africa
- Human papillomavirus infections and associated cancers
 - Global & SSA Data
- A case study of Nigeria
- Recommendations and Conclusion

Appreciation

- **Havey Institute for Global Health, Northwestern University**
- **Center for Global Oncology**
- **Special recognition**
 - Professor Rob Murphy
 - Professor Lifang Hou
 - Professor Isaac Adewole

Nigeria – is highly heterogeneous in culture, language, belief and social systems

- Huge population
- High poverty level
- Low literacy level
- Below average health outcomes
- High infection rates
- Considerable natural resources
- Conflicts
- Unstable polity
- Brilliant and hardworking

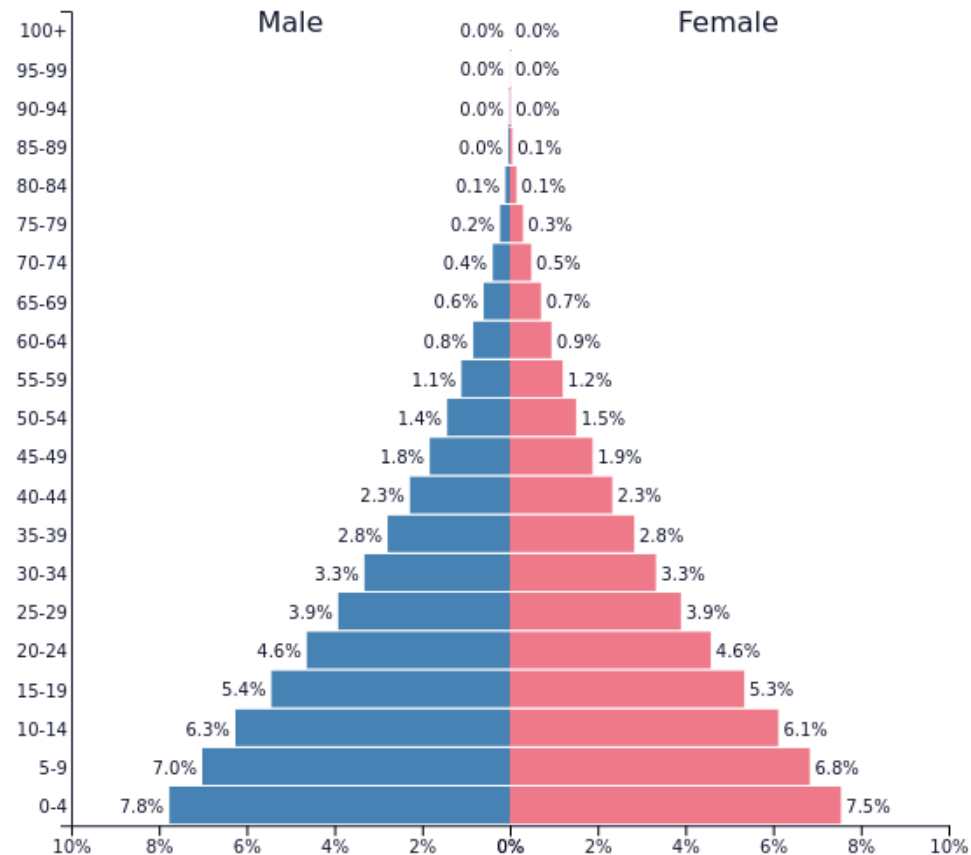


Various ethnic groups in Nigeria

The country has over 50 languages and over 250 dialects and ethnic groups. The three largest ethnic groups are the Hausa-Fulanis who are predominant in the north, the Igbos who are predominant in the south-east, and the Yorubas who are predominant in the south-west.

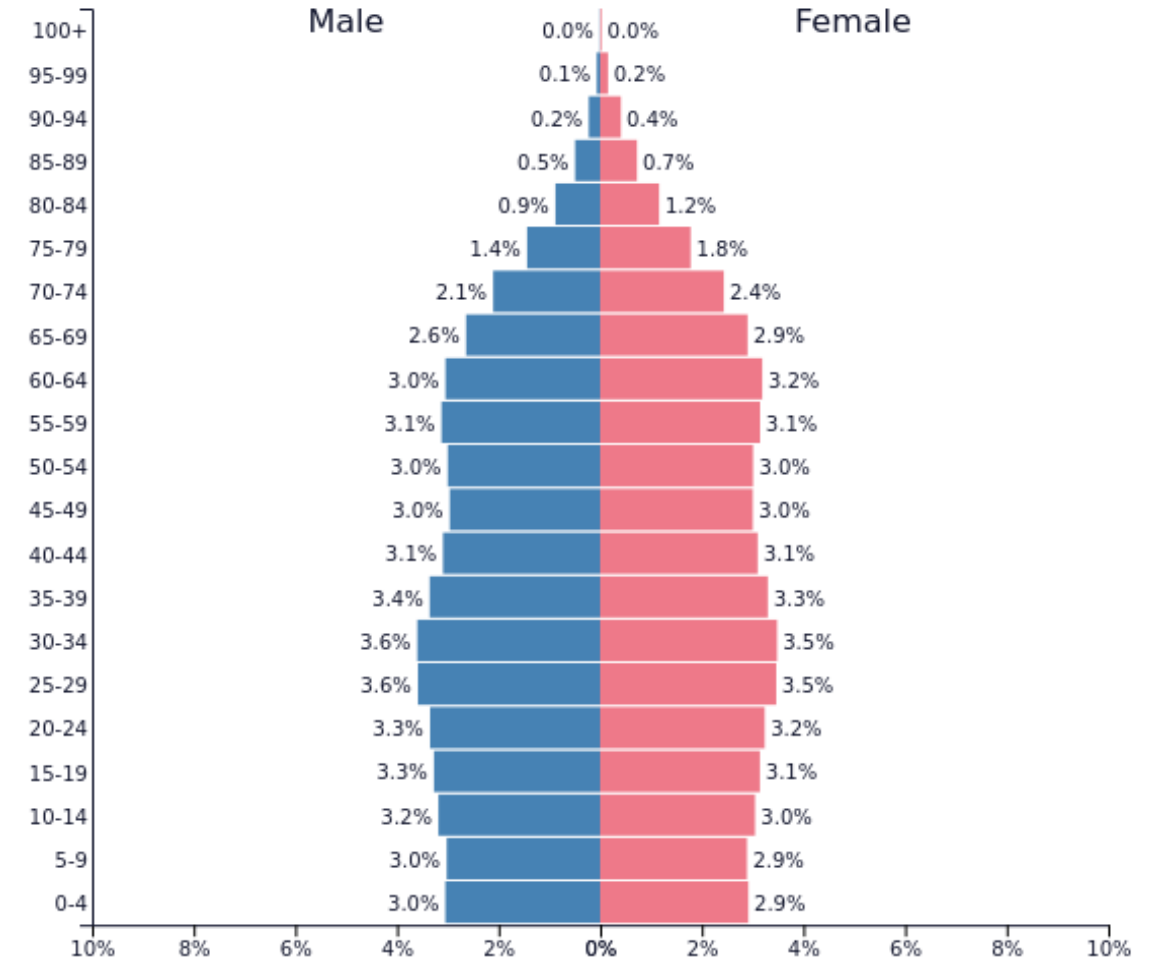


SSA has huge population of young people relative to United States



Sub-Saharan Africa - 2022
Population: **1,151,743,579**

PopulationPyramid.net



United States of America - 2022
Population: **334,805,268**

PopulationPyramid.net

What is a sexual act?

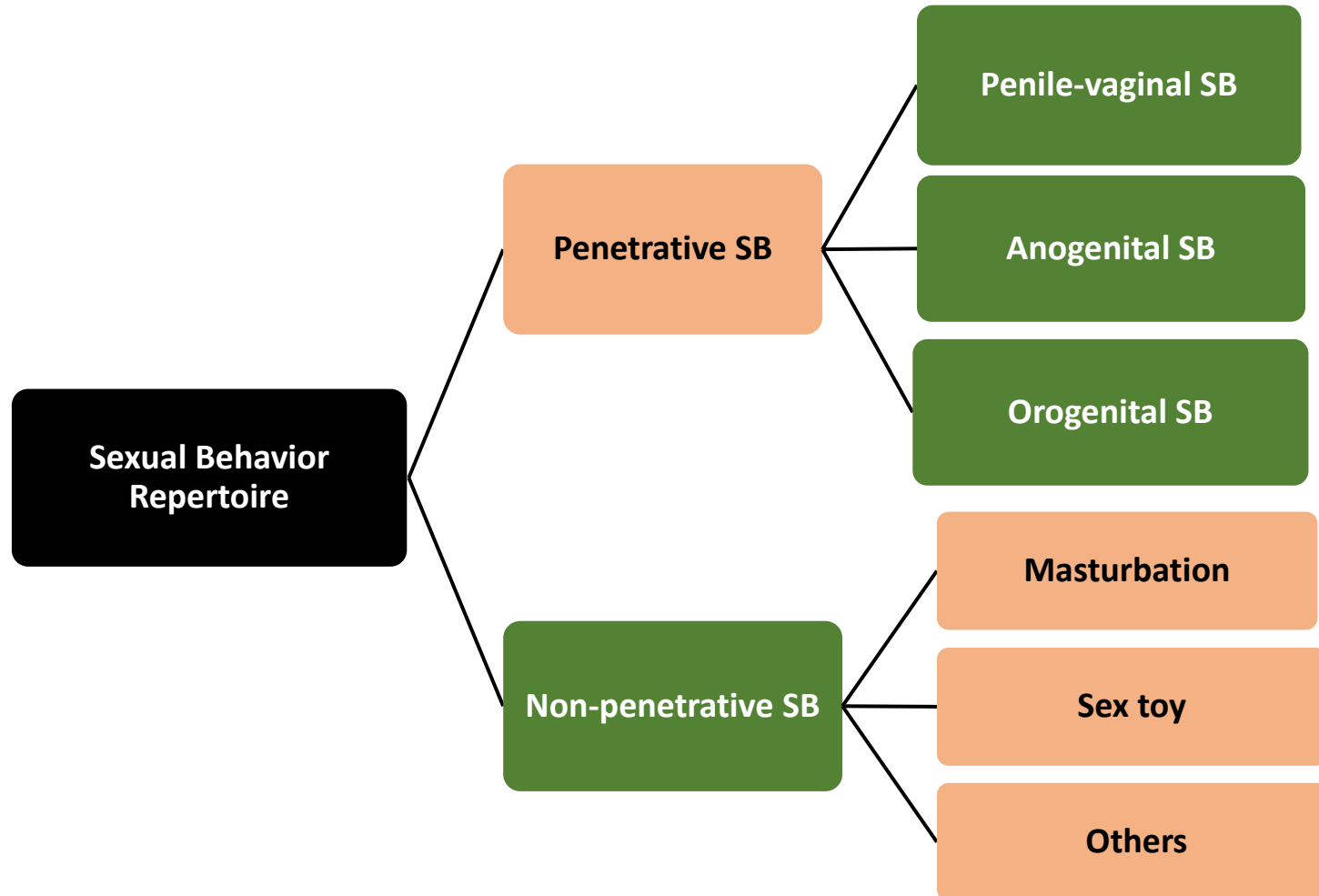
- **SEXUAL ACT**

- Contact between two or more persons by penetration of the **penis into the vagina or anus**
- Contact between the mouth and genitalia or by contact between the genitalia of one person and the genitalia or anus of another person
- Contact between the finger or hand of one person and the genitalia or anus of another person
- Using artificial sexual organs or substitutes in contact with the genitalia or anus

- **SEXUAL TOUCHING**

- Any contact with the breasts, buttock, groin, or genitals, or touching another with any of these body parts, or making another touch you or themselves with or on any of these body parts for the purpose of arousing or satisfying their own sexual desires

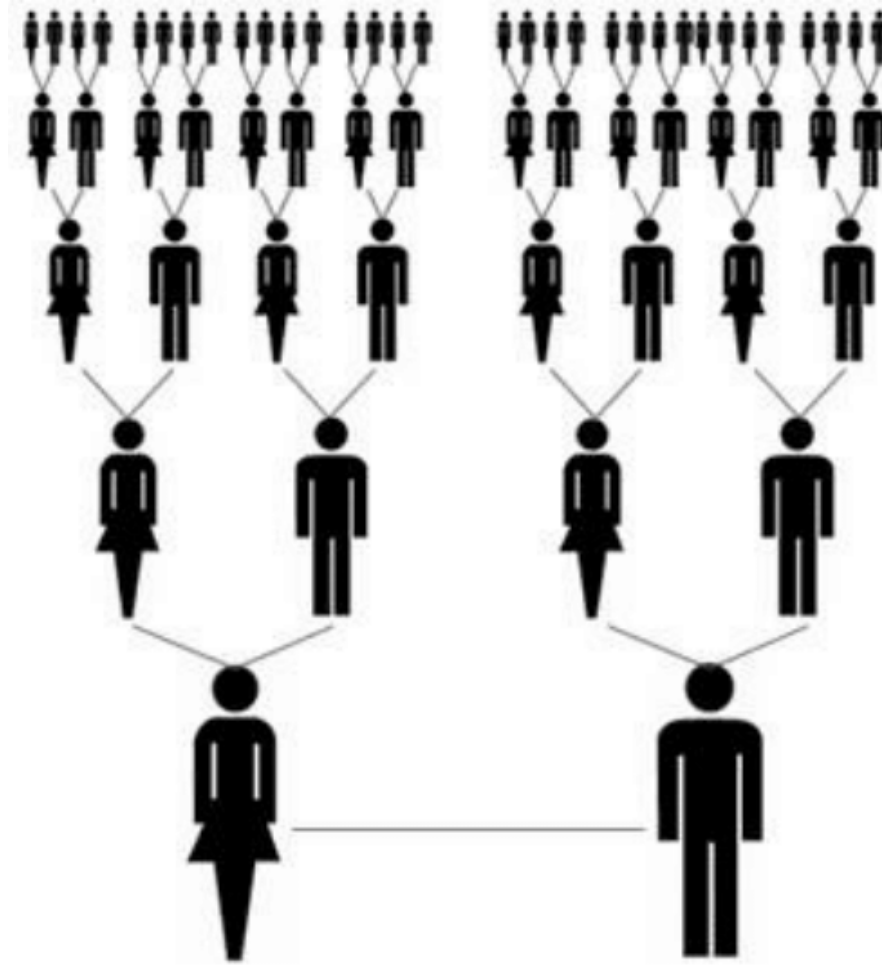
Sexual repertoire





Penetrative sexual risk behaviors

- **Condomless sex** (Barebacking) - i.e. sex without a condom
 - Penile-vaginal sex >85% of publications in SSA focused on PVS
 - Anogenital sex
 - Oro-genital sex
- **Other important attributes**
 - Early sexual debut
 - Multiple sexual partners
 - Having a high-risk partner - someone who has multiple sex partners or infections
 - Sex with a partner who has ever injected drugs
 - Engaging in sex work/transactional sex



When you have sex with someone, you are having sex
With everyone they have had sex with, and with
Everyone they have had sex with, and with
Everyone they have had sex with, and with
Everyone they have had sex with, and with
Everyone they have had sex with, and with
Everyone they have had sex with, and with
Everyone they have had sex with, and with



Changing sexual behaviors in young people is global – HICs – NATSAL DATA in UK

The proportion of sexually active 16-24 year olds who said they have had vaginal, oral and anal sex during the previous year

1990-1991

One in ten **women**

One in ten **men**



2010-2012

One in five **women**

One in four **men**



Data from National Surveys of Sexual Attitudes and Lifestyles (Natsal)

Sexual behaviour and Changing HPV burden

- **Penetrative** (Vaginal, oral and anal) and **non-penetrative** (self or mutual masturbation) sexual behaviours have been associated with HPV epidemiology
- In SSA a recent systematic review showed that adolescents and adults are increasingly engaging in unprotected vaginal, oral and anal sex

REVIEW

Open Access



Reported oral and anal sex among adolescents and adults reporting heterosexual sex in sub-Saharan Africa: a systematic review

Imran O. Morhason-Bello^{1,2*} , Severin Kabakama³, Kathy Baisley⁴, Suzanna C. Francis⁴ and Deborah Watson-Jones^{1,3}

Abstract

Background: Oral and anal sexual behaviours are increasingly reported among adolescents and adults reporting heterosexual sex in peer-reviewed journals in high income countries, but less is known about these behaviours in low and middle-income countries, especially in sub-Saharan Africa. The aim of this systematic review is to describe the prevalence of, and motivations for, oral and anal sex among adolescents and adults reporting heterosexual sex in sub-Saharan Africa.

Methods: A systematic review of published articles that reported oral and or anal sex in sub-Saharan Africa was conducted from seven databases up to and including 30th August 2018.

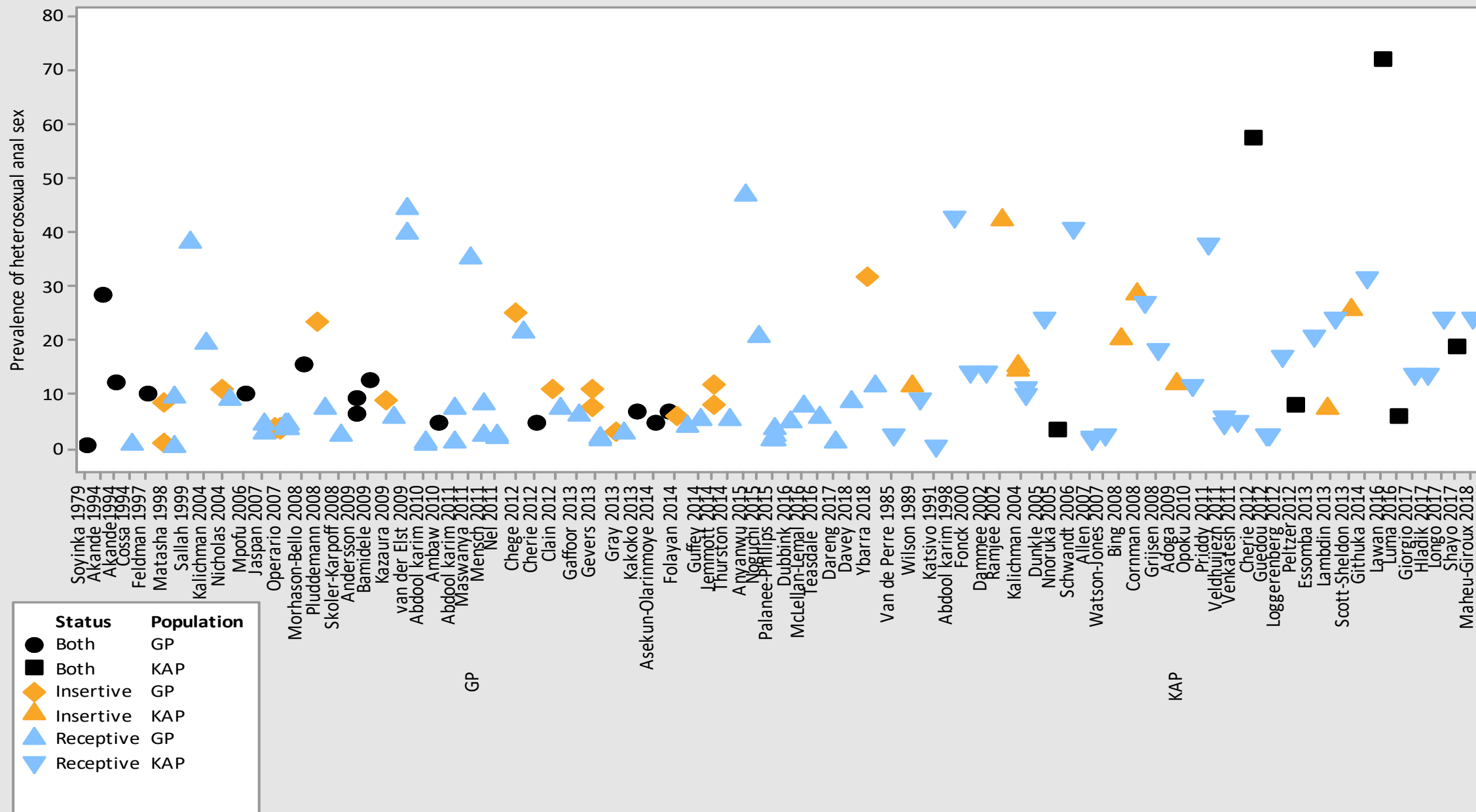
Results: Of 13,592 articles, 103 met the inclusion criteria. The prevalence of reporting ever practising oral sex among adolescents, university students and a combined population of adolescents/adults ranged from 1.7–26.6%, 5.0–46.4% and 3.0–47.2% respectively. Similarly, prevalences of reported ever practising anal sex ranged from 6.4–12.4% among adolescents, 0.3–46.5% among university students and 4.3–37.8% amongst combined population of adolescents and adults. Higher prevalences of oral and anal sex were reported among populations at high-risk for sexually transmitted infections and HIV and university students and, in most studies, both behaviours were more commonly reported by males than females. Heterosexual oral and anal sexual acts were associated with some high-risk behaviours such as inconsistent condom use and multiple sexual partners.

Conclusion: Reported oral and anal sex between men and women are prevalent behaviours in sub-Saharan Africa. Health professionals and policy makers should be aware of these behaviours and their potential associated health risks.

Keywords: Oral/anal sex, Sexual behaviour, Heterosexual, Adolescent, Adult, Sub-Saharan Africa

Only 4 out of 103 studies gave operational definition of oral and anal sex as well as information on risk identification

Limited or no information on health risks especially in West Africa



RESEARCH ARTICLE

Heterosexual oral and anal sex in Kinshasa (D.R.Congo): Data from OKAPI prospective cohort

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Heterosexual oral and anal sex in the Democratic Republic of Congo

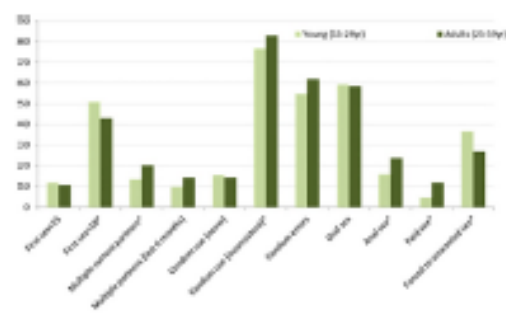
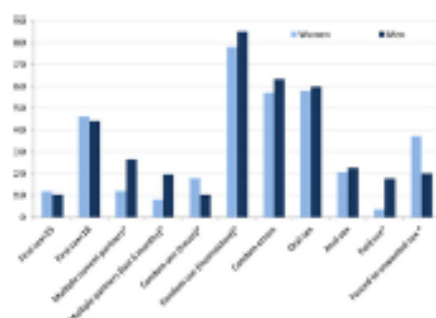


Table 1. Factors associated with oral and anal sex among the sexually active OKAPI cohort participants (N = 718).

	ORAL SEX			ANAL SEX		
	No (N = 296) (%)	Yes (N = 422) (%)	Crude OR (95%CI)	No (N = 563) (%)	Yes (N = 155) (%)	Crude OR (95%CI)
Sex: women vs. men	57.8	55.9	0.9 (0.7–1.2)	57.4	54.2	0.9 (0.6–1.3)
Age (yrs.): 25–49 vs. 15–24	72.3	71.6	1.0 (0.7–1.3)	69.8	79.3	1.7 (1.1–2.5)
Marital-monogamous: Yes vs. other	13.8	6.2	0.4 (0.2–0.7)	9.1	10.3	1.2 (0.7–2.2)
Media use						
The internet (daily): Yes vs. no	39.5	53.1	1.7 (1.3–2.3)	47.1	49.0	1.1 (0.8–1.5)
Mobile phone (daily): Yes vs. no	87.8	96.2	3.5 (1.9–6.5)	92.9	92.3	0.9 (0.5–1.8)
Perceived low community HIV risk: Yes vs. no	82.1	89.6	1.9 (1.2–2.9)	86.5	86.4	1.0 (0.6–1.7)
Alcohol consumption (daily): Yes vs. no			2.7 (1.3–5.5)			2.5 (1.3–4.7)
Age at first sex (yrs.): Yes vs. no	42.6	47.2	0.9 (0.9–0.9)	41.6	58.7	0.9 (0.9–1.0)
Ever pregnant (women): Yes vs. no	42.7	61.0	1.6 (1.1–2.2)	48.9	70.2	1.6 (1.1–2.3)
Multiple concurrent sexual partners: Yes vs. no	9.8	24.2	2.9 (1.9–4.6)	14.9	30.3	2.5 (1.6–3.7)
Condom use: inconsistent vs. correct	68.0	89.8	6.2 (3.7–10.3)	76.0	93.0	3.8 (2.0–7.3)
Type of sex						
Anal sex: Yes vs. no	NA	30.6	4.6 (2.9–7.2)	NA	NA	NA
Oral sex: Yes vs. no	NA	NA	NA	52.0	83.2	4.6 (2.9–7.2)
Paid sex: Yes vs. no	4.0	13.7	3.8 (2.0–7.2)	6.6	21.3	3.8 (2.3–6.4)
Ever partner-forced to unwanted sex: Yes vs. no	21.3	35.5	2.0 (1.4–2.9)	26.8	40.0	1.8 (1.2–2.6)

perception of community risk, alcohol consumption, age at first sex, pregnancy, multiple partners, condom use, anal sex, paid sex and forced sex.

* Logistic regression model including the following variables: sex, age, alcohol consumption, early sex, pregnancy, multiple partners, condom use, oral sex, paid sex and forced sex.

NA: Not applicable.

RESEARCH ARTICLE

Oral sex practices among men who have sex with men and transgender women at risk for and living with HIV in Nigeria

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Table 3. Association of oral sex and bacterial oropharyngeal STIs among MSM and TGW in Nigeria.

STIs	Any Oral Sex*	No Oral Sex	Crude OR**
	n (col%)	n (col%)	(95% CI)
	n = 500	n = 252	
Any Oropharyngeal STI	43 (8.6)	9 (3.6)	2.5 (1.2, 5.3)
Oropharyngeal Gonorrhea	31 (6.2)	8 (3.2)	2.0 (0.9, 4.4)
Oropharyngeal Chlamydia	15 (3.0)	2 (0.8)	3.9 (0.8, 1.1)

Abbreviations: STI, sexually transmitted infection; col, column; OR, odds ratio; CI, confidence interval; MSM, men who have sex with men; TGW, transgender women.

*Oropharyngeal STIs categorized as any oral *Neisseria gonorrhoeae* [NG] and/or *Chlamydia trachomatis* [CT].

Unadjusted odds ratios calculated using logistic regression model. **Bolded indicates $p < 0.05$.

Table 2. Demographic and behavioral characteristics associated with oral sex among MSM and TGW in Nigeria.

Factors	Oral sex (n, row %) n = 920	No oral sex (n, row %) n = 422	OR (95% CI)	aOR [†] (95% CI)
HIV status				
At risk	326 (59)	230 (41)	Ref.	Ref.
Living with	304 (75)	192 (25)	2.1 (1.6–2.6)	1.4 (1.1–1.8)
Gender identity				
Man	322 (66)	349 (34)	Ref.	Ref.
Woman	122 (32)	27 (18)	2.1 (1.4–3.3)	1.8 (1.1–2.8)
Non-binary	76 (75)	26 (25)	1.4 (0.9–2.3)	1.2 (0.7–2.0)
Mobile phone owned				
No	39 (48)	31 (52)	Ref.	Ref.
Yes	491 (69)	391 (31)	2.4 (1.5–4.1)	2.3 (1.3–3.9)
Alcohol use ^{**}				N/A
No	427 (64)	230 (36)	Ref.	
Yes	491 (74)	192 (26)	1.6 (1.3–2.0)	
Condom use anal sex ^{***}				N/A
No	308 (73)	126 (27)	Ref.	
Yes	378 (66)	296 (34)	0.8 (0.6–1.0)	
Prior HIV test				Ref. 1.6 (1.2–2.2)
No	159 (60)	107 (40)	Ref.	
Yes	361 (71)	315 (29)	1.6 (1.2–2.1)	
Receptive anal sex [†]				
No	176 (53)	156 (47)	Ref.	Ref.
Yes	364 (74)	266 (26)	2.4 (1.9–3.1)	1.7 (1.3–2.3)
Insertive anal sex [†]				N/A
No	307 (71)	86 (29)	Ref.	
Yes	311 (68)	336 (32)	0.9 (0.7–1.2)	
Male anal sexual partners [†]				
0–1	46 (50)	67 (50)	Ref.	Ref.
2–4	307 (62)	192 (38)	1.6 (1.1–2.3)	1.5 (1.0–2.2)
5+	365 (77)	163 (23)	3.3 (2.3–4.8)	2.9 (1.9–4.3)
Female vaginal sexual partners [†]				
0–1	458 (72)	235 (28)	Ref.	Ref.
≥2	362 (61)	167 (39)	0.6 (0.5–0.8)	0.7 (0.5–0.9)
Transaction [†]				
No	466 (66)	246 (34)	Ref.	Ref.
Yes	454 (72)	176 (28)	1.3 (1.1–1.7)	1.2 (0.9–1.5)

Abbreviation: OR, odds ratio; CI, confidence interval; MSM, men who have sex with men; TGW, transgender women.

[†]Multivariable logistic regression model was adjusted for gender identity, mobile phone, prior HIV test, receptive anal sex male partners, and transaction. Alcohol, condom use anal sex, insertive anal sex were not determined to be significant confounders and were excluded from the model. Bolded indicates $p < 0.05$.

^{**}Sexual activity and partner variables assessed in the past 12 months.

^{***}Alcohol use assessed in the past 30 days.

^{††}Condom use reported for most recent anal sex act with a male partner.



HHS Public Access

Author manuscript

AIDS Behav. Author manuscript; available in PMC 2017 April 01.

Published in final edited form as:

AIDS Behav. 2016 April ; 20(4): 754–762. doi:10.1007/s10461-015-1120-0.

Condoms, Lubricants and Rectal Cleansing: Practices Associated with Heterosexual Penile-Anal Intercourse Amongst Participants in an HIV Prevention Trial in South Africa, Uganda and Zimbabwe

Zoe Duby^{1,2,6}, Miriam Hartmann³, Elizabeth T. Montgomery³, Christopher J. Colvin², Barbara Mensch⁴, and Ariane van der Straten^{3,5}

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- Poor knowledge of PAI
- Do not know that PAI has health risks
- Engage in dry anal sex
- Mostly condomless



Abstract

We investigated condom and lubricant use, rectal cleansing and rectal gel use for penile-anal intercourse (PAI), during in-depth interviews with women from South Africa, Uganda and Zimbabwe who formerly participated in VOICE, a five-arm HIV prevention trial of two antiretroviral tablets and a vaginal gel. Few studies have addressed practices related to PAI among women; existing data from Africa on condom and lubricant use for PAI, as well as preparatory practices of PAI such as rectal cleansing, are limited to men who have sex with men. Women demonstrated a lack of awareness of HIV transmission risks of PAI and none of the participants reported using condom-compatible lubricants for PAI. Participants described a variety of preparatory rectal cleansing practices. Some participants disclosed rectal use of the vaginal study gel. Understanding practices related to PAI in Africa is critical to microbicide development, as these practices are likely to influence the acceptability, feasibility, and use of both vaginal and rectal microbicide products.

Keywords

Condom; Lubricant; Rectal cleansing; Heterosexual; Anal intercourse; HIV

Human papillomavirus Infection

HPV infection transmission

- Chiefly transmitted through unprotected vaginal, oral and anal sex
- It is also transmitted through non-penetrative sexual behaviours, skin-to-skin contact and vertically from mother to child
- Transmission is usually via breach of epithelial surface

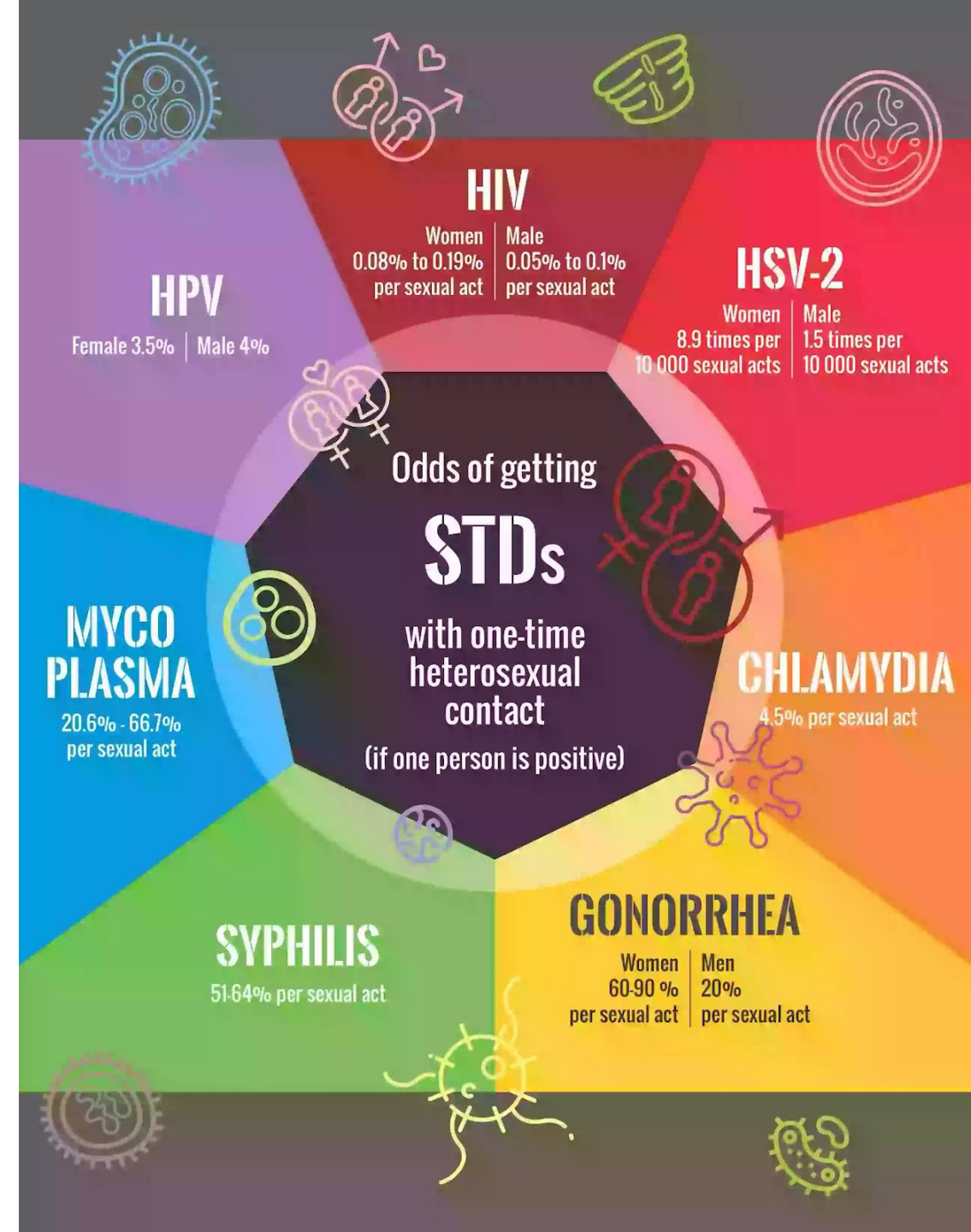
Current Classification of HPV Genotypes by IARC [2009]

Classifications	HPV Genotypes
1	16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58 and 59
2A	68
2B	26, 30, 34, 53, 55, 66, 67, 69, 70, 71, 72, 73, 81, 82, 83, 84, IS39 and CP6108
3	6, 11

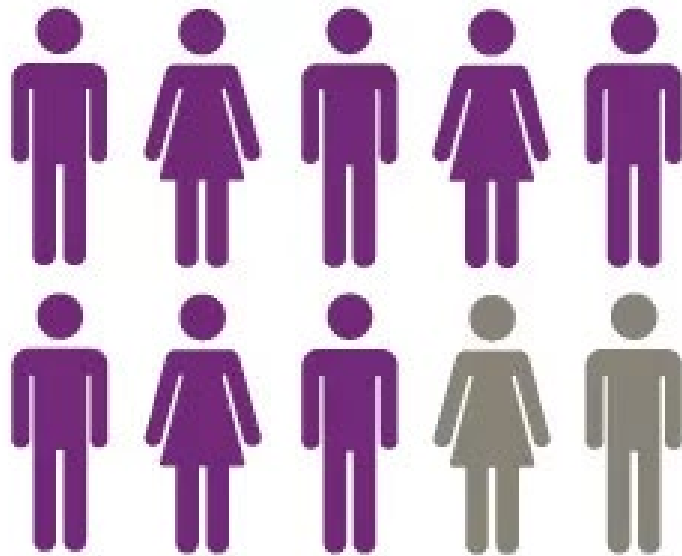
Class 1 & 2A are High-risk; Class 2B & 3 are Low-risk

HPV risk relative to other STIs

- In general, HPV is more easily acquired and transmitted relative to HIV infection per sexual contact
- Irrespective of other potential discriminatory factors
 - Sexual behaviors & associated factors
 - Lifestyle
 - Biological factors



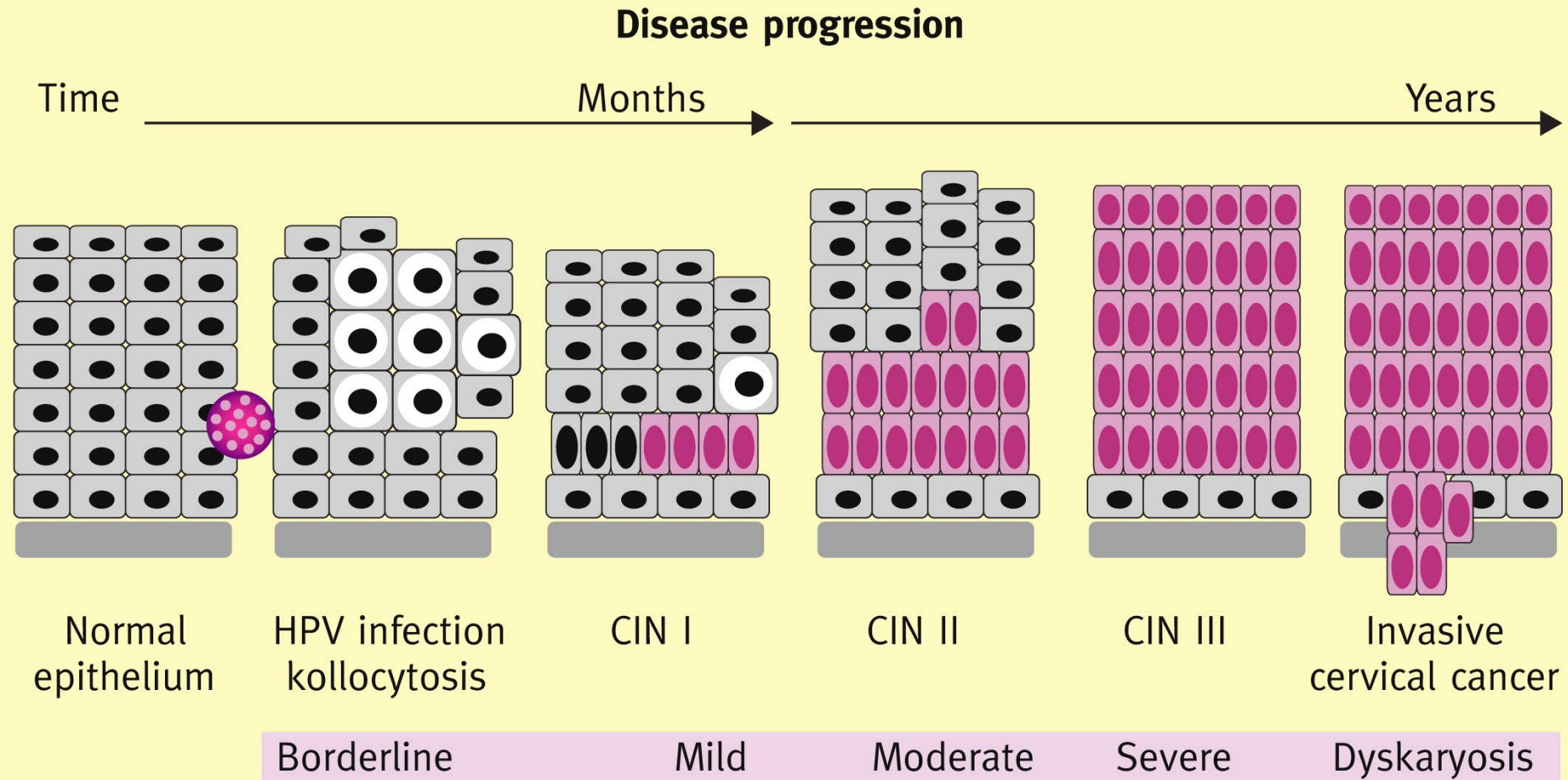
**HPV is a common virus that infects
teens and adults.**



80%

**of people will get an HPV
infection in their lifetime.**

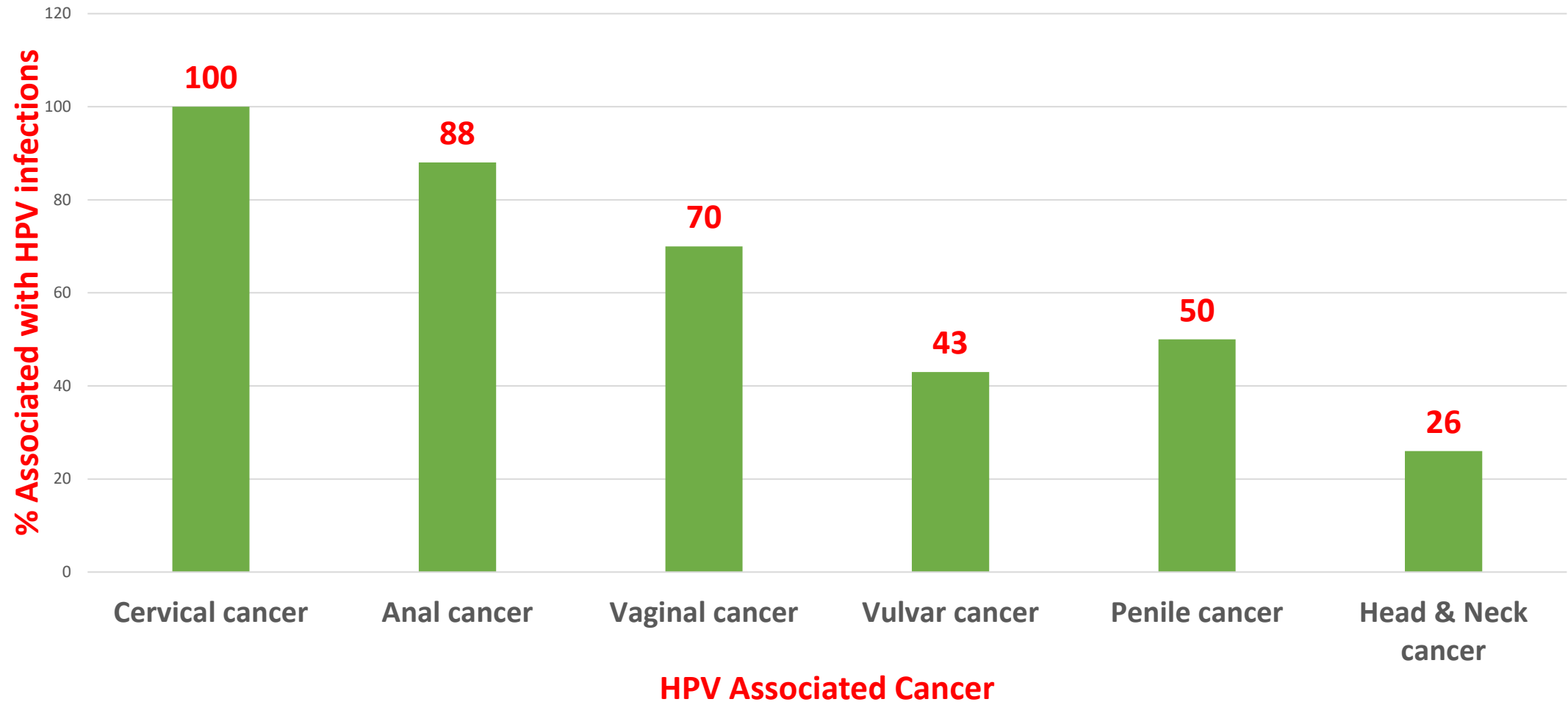
Natural history of HPV infection and disease progression



CIN I 57% CIN II 43% CIN III 32%

Approximate likelihood of regression

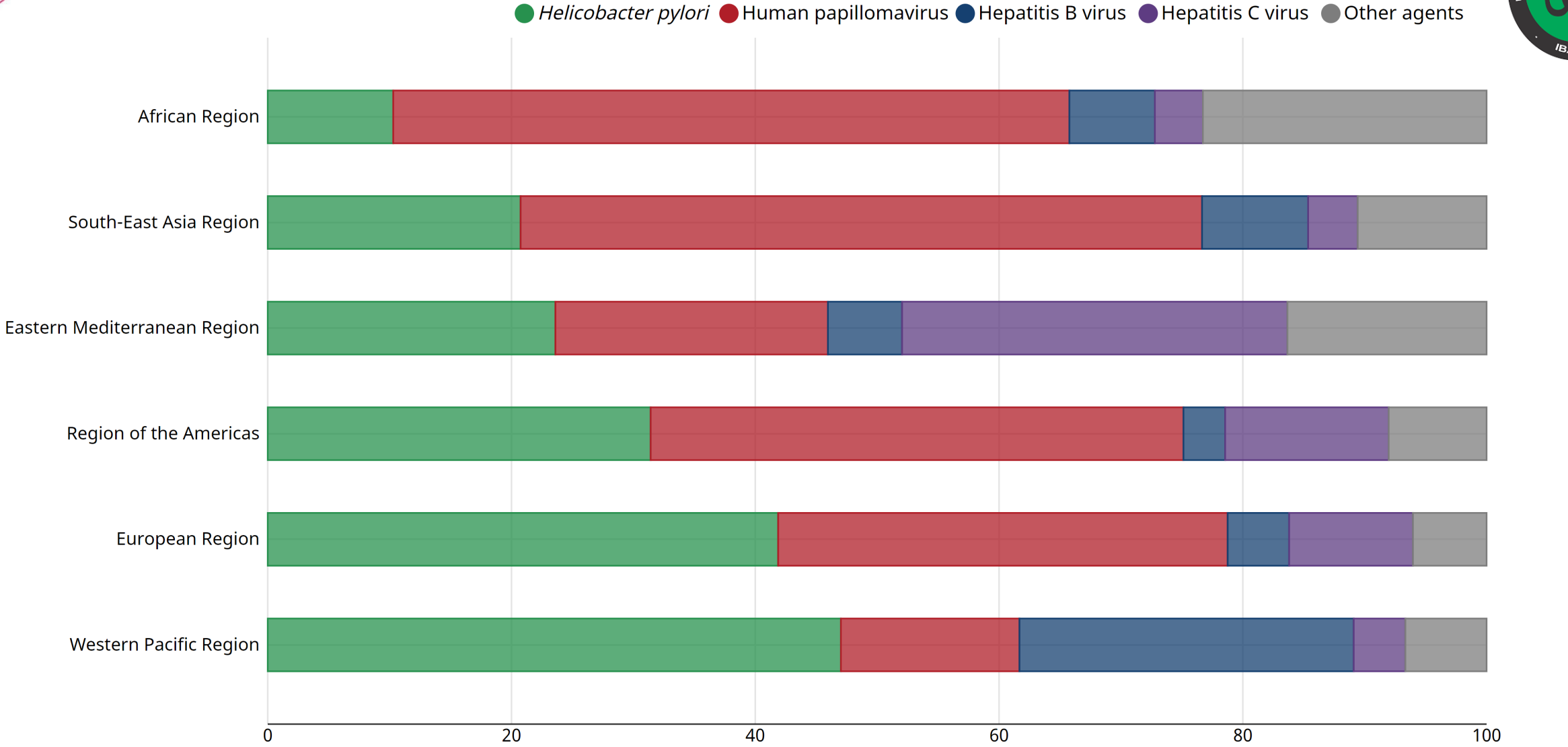
HPV Associated Cancers

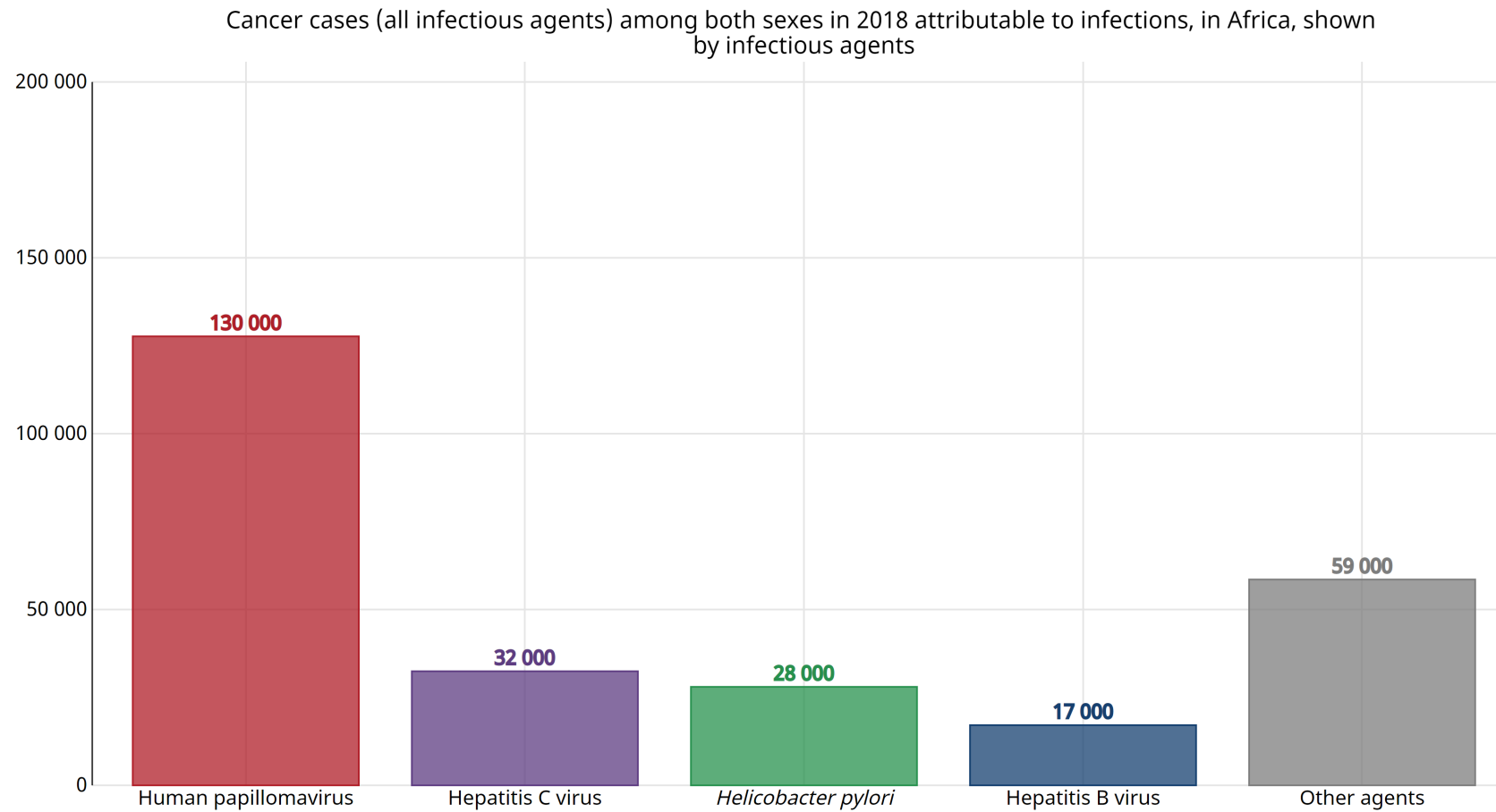


Cancer cases (all infectious agents) among both sexes in 2018 attributable to infections, in the world, shown by infectious agents



Proportions of cancer cases among both sexes attributable to infections in 2018 (by WHO region)

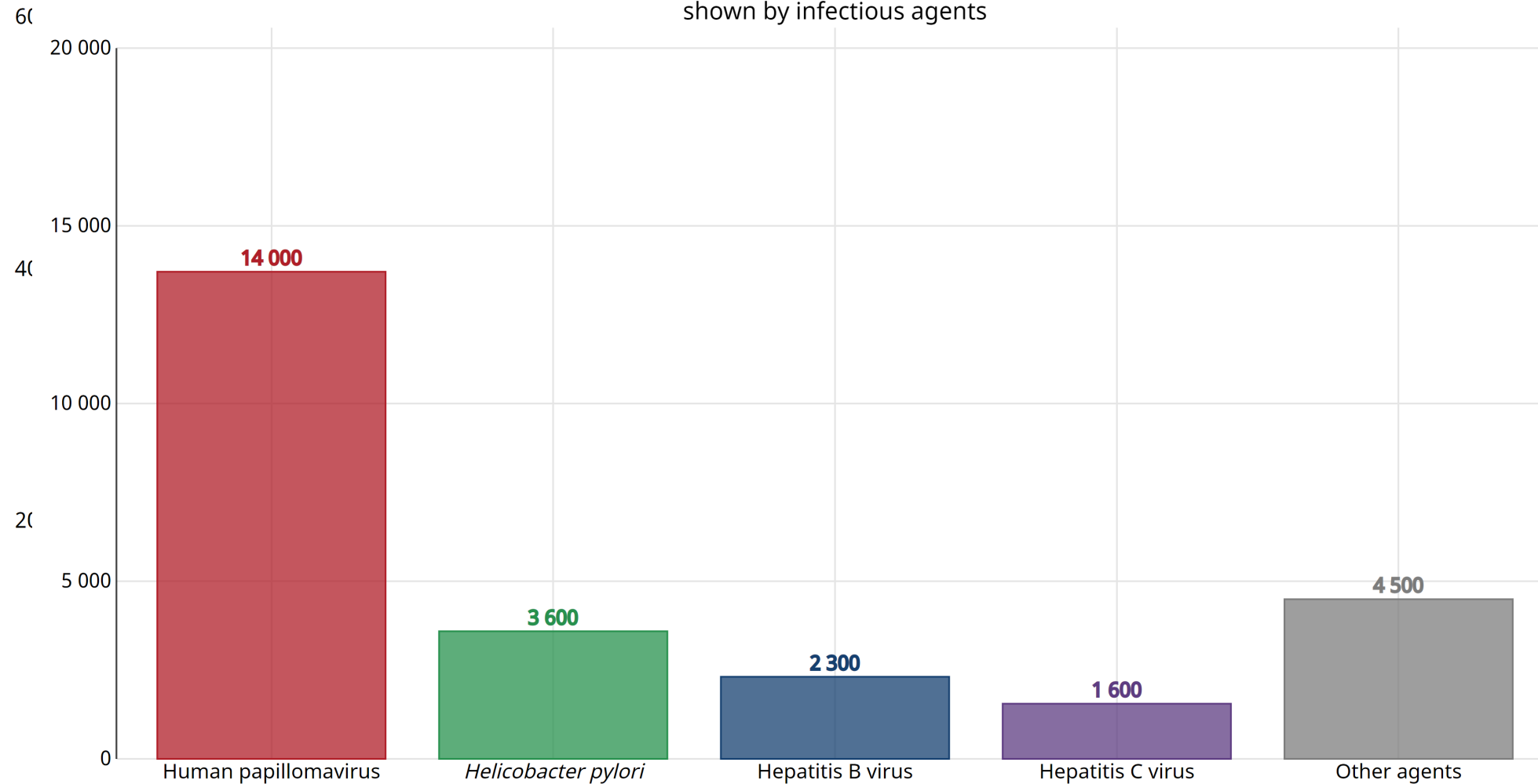






Cancer cases (all infectious agents) among both sexes in 2018 attributable to infections, in Southern Africa,

Cancer cases (all infectious agents) among both sexes in 2018 attributable to infections, in Central Africa,
shown by infectious agents



Data source: de Martel C, Georges D, Bray F, Ferlay J, Clifford GM (2020)
Graph production: Global Cancer Observatory (<http://gco.iarc.fr/>)
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International Agency for Research on Cancer



Research on Cancer



Cancer in Africa 2018: The role of infections

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Table 2. Estimated cancers attributable to Human Papillomaviruses in Africa in 2018

Cancer (PAF %)	Sub-Saharan Africa				All Africa					
	Male	(% total cancer)	Female	(% total cancer)	Male	(% total cancer)	Female	(% total cancer)	Both	(% total cancer)
Cervix (100%)			111,630	24.3%			119,280	19.6%	119,280	11.3%
Anus - SCC only (80–90%) ¹	620	0.20%	1,490	0.32%	840	0.19%	1,750	0.29%	2,590	0.3%
Vulva (40%) ²			2040	0.44%			2,210	0.36%	2,210	0.2%
Vagina (72%)			1,230	0.27%			1,420	0.23%	1,420	0.1%
Penis (40%)	890	0.29%			910	0.20%			910	0.1%
Oro-pharynx (30%)	430	0.14%	210	0.05%	510	0.11%	250	0.04%	760	0.1%
Oral cavity (4%)	230	0.07%	190	0.04%	300	0.07%	250	0.04%	550	0.1%
Larynx (4%)	200	0.06%	50	0.01%	340	0.08%	70	0.01%	410	0.0%
(Head & Neck)	(860	0.27%)	(450	0.10%)	(1,150	0.26%)	(570	0.09%)	(1,720	0.2%)
All HPV related	2,370	0.76%	116,840	25.4%	2,900	0.65%	125,230	20.6%	128,130	12.1%

¹SCC cases: 40% (Males)–60% (Females) of anal cancers in SSA; 60% (M) and 70% (F) of anal cancers in NA.

²PAF is age related (15–54: 48%; 55–64: 28%; 65+: 15%).

The burden of oral cancer in sub-Saharan Africa: An estimate as presented to the Global Oral Cancer Forum, March 2016

Jos Hille¹ and Newell W Johnson²

Table 1. African regional ASIRs and mortality rates per 100,000 pa for lip and intra-oral cancer (C00–C06) in males and females (IARC/ GLOBOCAN 2012).

Region	Males		Females		All		M–F ratio	
	ASIR	Mortality	ASIR	Mortality	ASIR	Mortality	ASIR	Mortality
World	5.5	2.7	2.5	1.2	4	1.9	2.2:1	2.3:1
WHO AFRO	3.4	2.2	2	1.4	2.7	1.8	1.7:1	1:1.6
Western Africa	1.7	1.2	1.4	1.0	1.5	1.1	1.2:1	1:1.6
Eastern Africa	4.5	3.2	2.8	1.9	1.5	2.5	1.6:1	1:1.7
Southern Africa	6.3	2.8	2.3	1.0	4	1.7	2.7:1	1:1
Sub-Saharan Africa	3.5	2.4	2.1	1.4	2.7	1.9	1.7:1	1.7:1

WHO: World Health Organization; IARC: International Agency for Research on Cancer.

Hille and Johnson

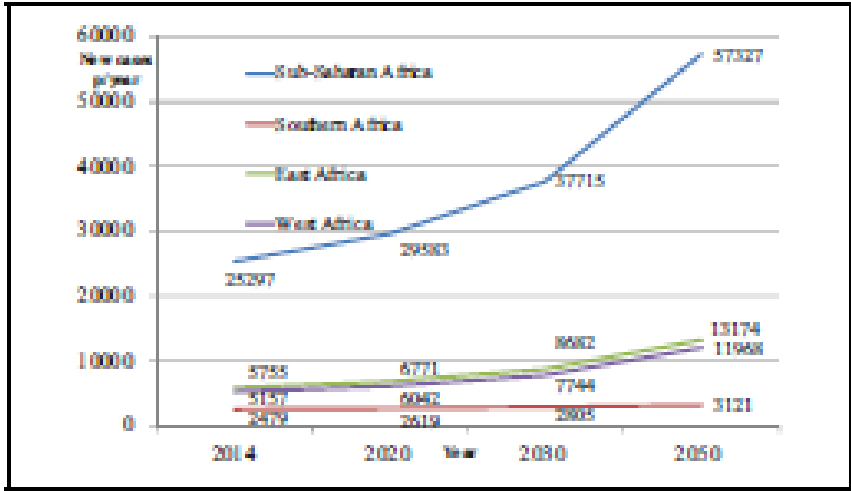


Figure 11. Forward projection of the minimum number of new intra-oral and lip cancer (C00–C06) cases for 2020, 2030 and 2050 for sub-Saharan Africa and three major regions. Source: GLOBOCAN 2012 and Atlas/United Nations population pyramids websites.





RESEARCH

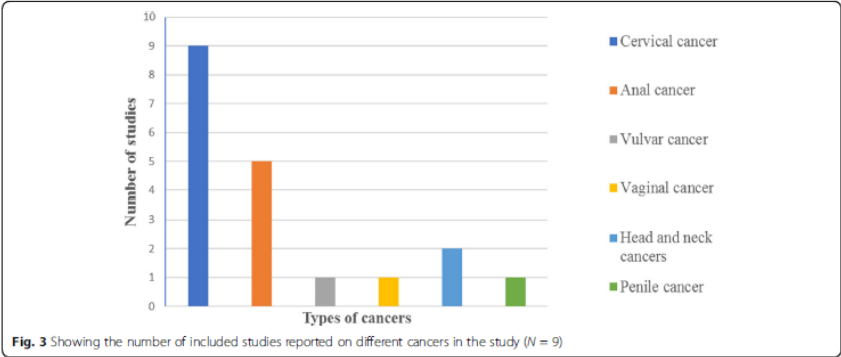
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The interplay of HIV and human papillomavirus-related cancers in sub-Saharan Africa: scoping review



Kabelo Matjie Bridget Lekoane¹, Desmond Kuupiel^{1,2*}, Tivani P. Mashamba-Thompson¹ and Themba G. Ginindza¹

Anatomic sites of HACs	No of Studies	Risk in PLWHA
Cervical cancer	9	2-22% ICC
Anal Cancer	5	2-10%
Vulvar & Vaginal cancer	1	? Younger
Head & Neck Cancer	2	2-4%
Penile Cancers	1	4-22%





Sexual Behaviours & HPV Infections in Nigerians in Ibadan (SHINI) Project





Rationale

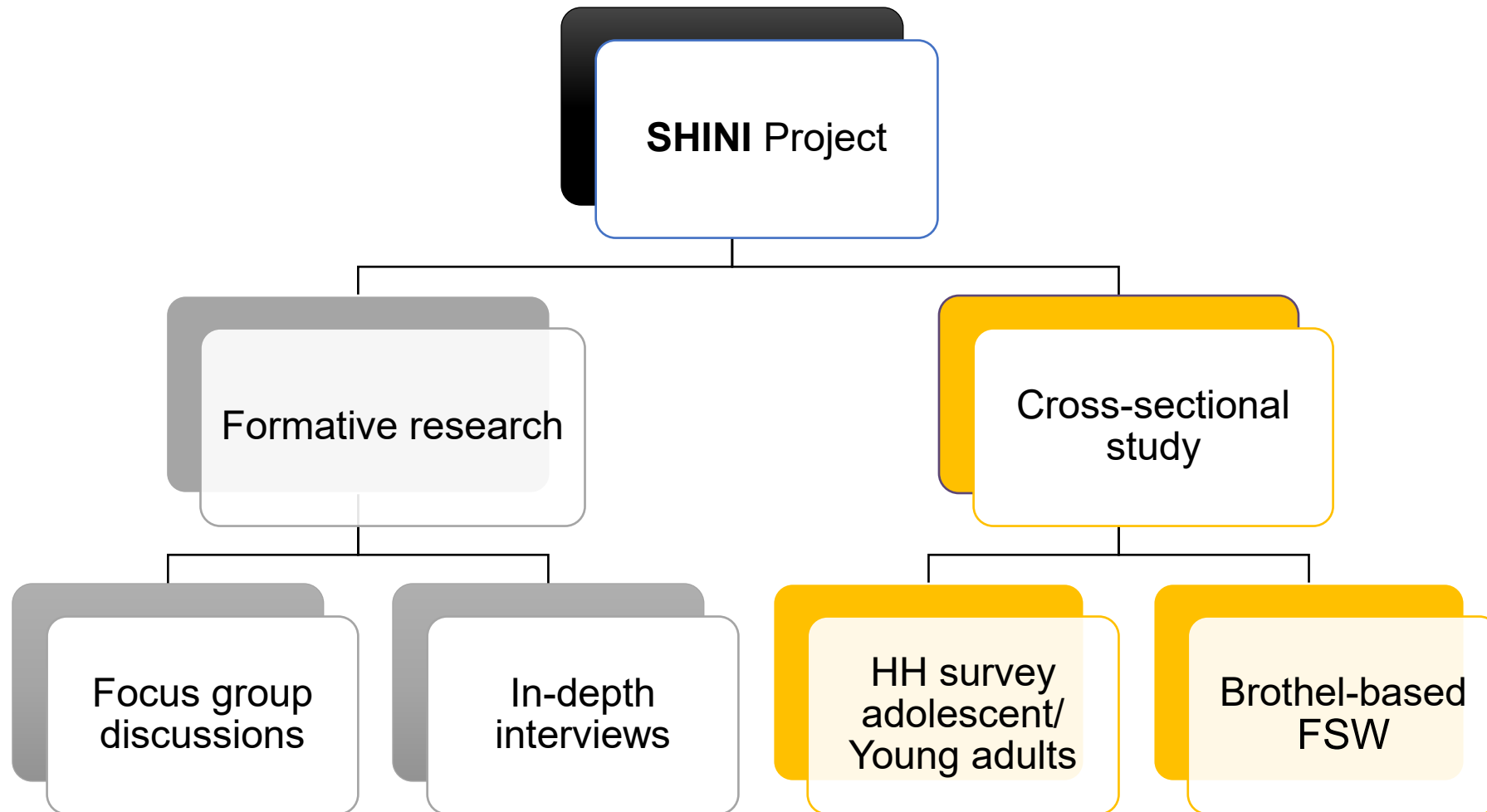
- Despite high prevalence of genital HPV in women in SSA, data on oral and anal HPV in men and women are limited
- There is also no data globally on prevalence of HPV infection in vulva of healthy women
- **In Nigeria**
 - The burden of cervical HPV infection is second to East Africa but there are no data on other anatomic sites in women (**20-50%**)
 - The 2015 ICO report shows that **14,089** incident cervical cancer cases and 8,240 related deaths occur each year in Nigeria accounting for **51.5%** and **49.8%** respectively of the total cases in West Africa
 - No existing data on HPV infection in men and high risk populations
 - The quality of information reported on oral and anal sex make it difficult to compare with similar studies from developed countries

Thomas, J.O., et al. Br J Cancer, 2004. **90**(3): p. 638-45. Ezechi OC, et al 2014. Virol J. 2014
Akarolo-Anthony, S.N., et al BMC Public Health, 2014. **14**: p. 656. Wagner, et al Infect Agent Cancer, **10**(13): p. 1-6. 2015;

Objectives of the study

1. Explore knowledge, socio-cultural interpretation and understand how sexually active adolescent/young adults, adults and FSW in Nigeria learn different sexual behaviours.
2. Describe attitude, motivations, how potential health risks are understood and stigma for engaging in any of these sexual behaviour.
3. Determine the prevalence of oral, genital (penile/cervical/vulval) and anal HPV infections.
4. Determine the risk factors associated with any HPV infection in each of the anatomic sites.
5. Examine the concordance of HPV genotype-specific infections in different anatomical sites.

Sexual behaviour and HPV Infection in Nigerians in Ibadan (SHINI) Project



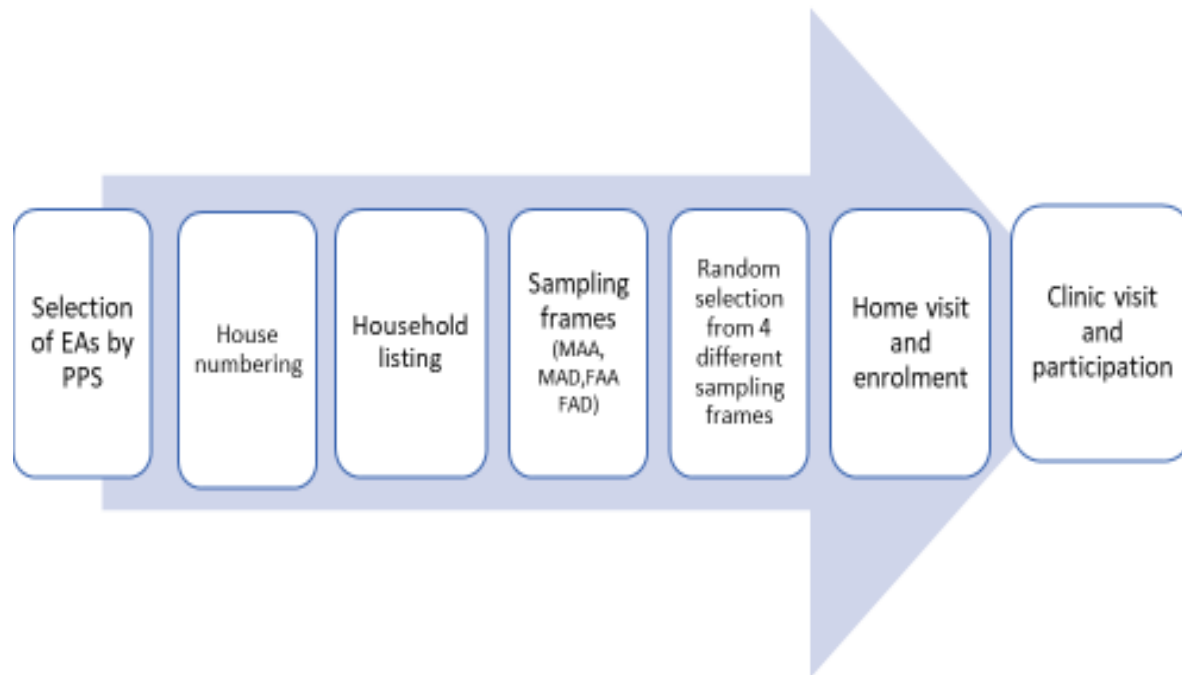


Material and Methods

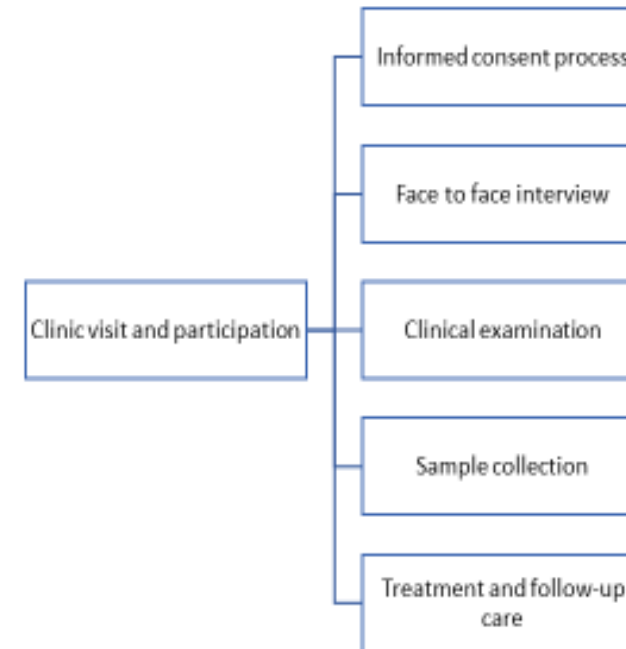
- A cross-sectional household study of hetero-sexually active men and women aged in two communities and brothel-based sex workers in Ibadan, Nigeria
- **A face-to-face interview, clinical examination and oral, cervical, vulvar, anal and penile samples were collected from participants**
- **Anyplex II 28HPV assay was used to test samples for 28 HPV genotypes**
- Descriptive and multivariable analyses were used to report prevalence and risk factors associated with HPV infection
 - Explanatory variables: Sociodemographic, Behavioural and Biological Factors
 - We used hierarchical modelling
- Analyses were performed with STATA 16.0 software

Summary of study procedure

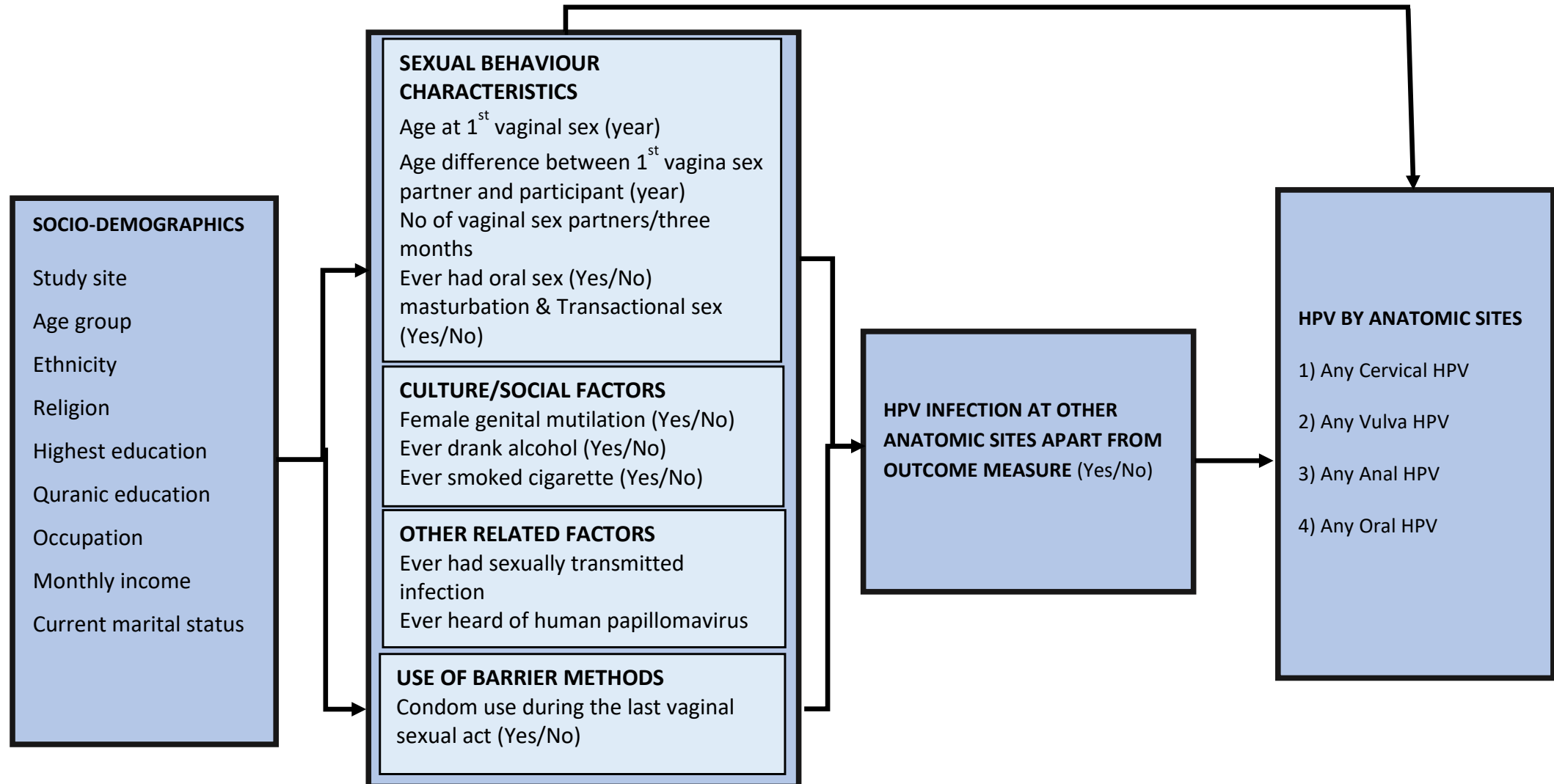
Flow of field activity



Clinic activities

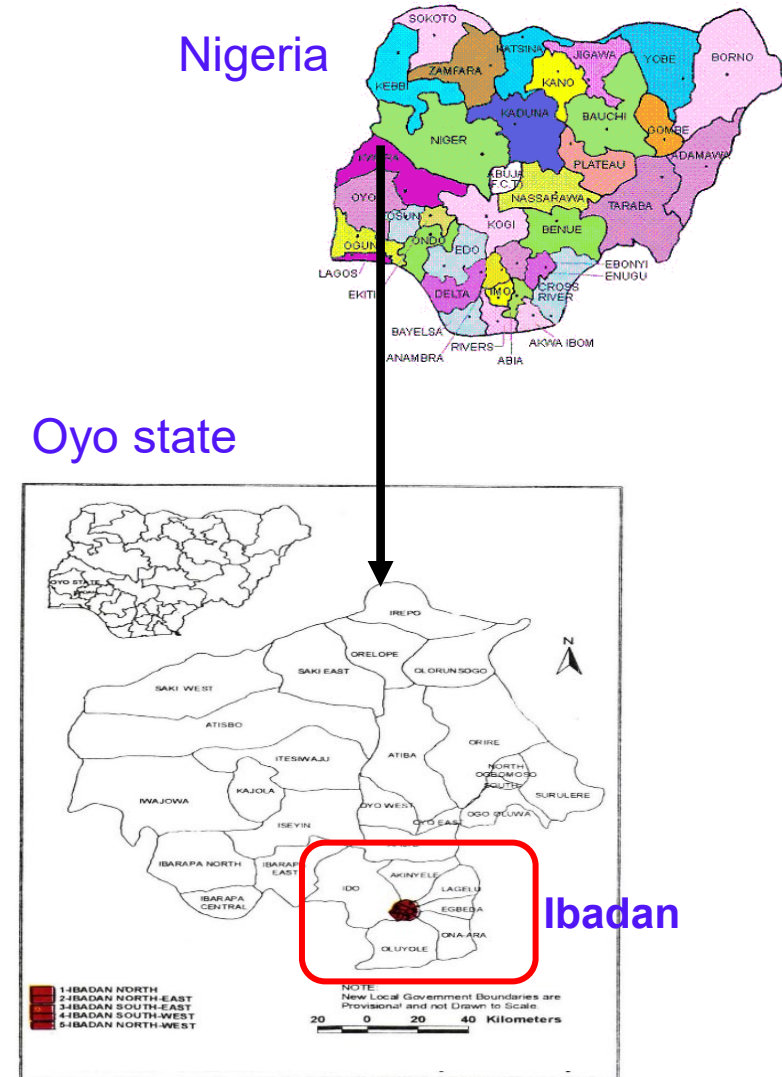


Conceptual Framework for the risk factor analysis of any HPV infection



Study site

- Nigeria has a population of about 211 million people
- **Ibadan metropolis**
 - Located in SW Nigeria
 - Local Population is 2.6 million
 - Administratively divided into 11 Government Areas (LGAs)
 - Demographically, there are 6 urban LGAs while the rest are rural/peri-urban mix



Qualitative Findings



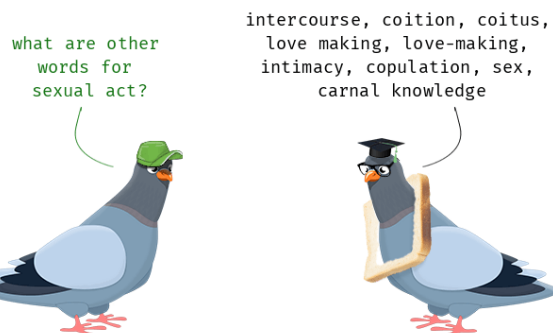
Heterosexual Oral and Anal Sex: Perceptions, Terminologies, and Attitudes of Younger and Older Adults in Ibadan, Nigeria

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Received: 14 October 2020 / Revised: 15 February 2022 / Accepted: 17 February 2022
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Abstract

Although heterosexual oral and anal sexual behaviors have been reported in sub-Saharan Africa, little is known about how they are understood and perceived, particularly, in West Africa. We undertook a qualitative exploration of local terminologies and sexual scripts associated with heterosexual oral and anal sex in preparation for a quantitative survey. We held focus group discussions (18) and interviews (44) with younger and middle-aged men and women from the general population and female sex workers (FSWs) in selected communities in Ibadan. Most participants had heard of oral and anal sex. Younger adults aged 18–25 years, particularly male participants and FSWs, appeared more informed than older adults in the general population. Sexually explicit movies were the most cited source of information. Oral and anal sexual behaviors were considered sensitive, with different local names, meanings, and interpretations. Participants advised against the use of slang terms in research. We identified six different scripts employed by participants in discussing oral and anal sex practices: protecting sexual relationship, financial reward, an alternative to vaginal sex, pleasure, male dominance and control, and risk, stigma, and disgust.



THE FOCUS OF THIS PAPER

- **Awareness/Knowledge & source of information**
- **Motivations for and against oral and anal sexual practice**
 - Protection of relationship;
 - financial reward; alternative to vaginal sex & pleasure
- **Local slang terms**
- **Attitude toward use of slang terms for research**

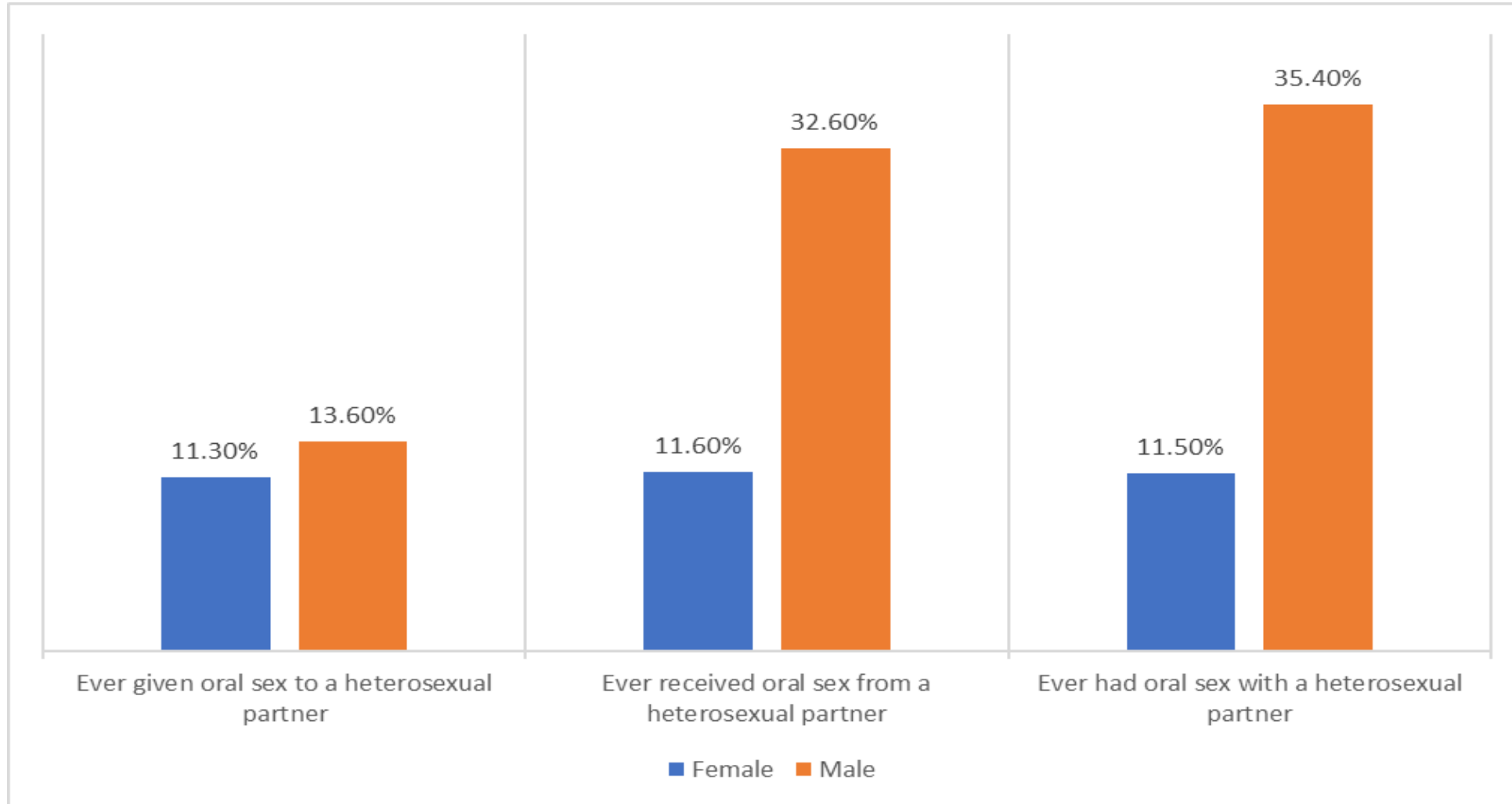
Table 3 Local names or slang terms for oral and penile–anal sex

Oral and penile–anal sex definitions	Examples of slang terms
Slang term when a man gives oral sex to a woman (i.e., when a man/boy uses his tongue, mouth, or lips to touch or stimulate the external genitals of a woman/girl)	'lick plate,' 'clean up,' 'wash wash,' and 'go south'
Slang term when a woman gives oral sex to a man or boy, i.e., when a woman uses her tongue or mouth to touch or lick or suck the penis	'blow job (BJ),' 'chop or eat banana or plantain ('dodo'), 'eating carrot,' 'hold microphone to sing,' 'licking soft sweet stick,' 'lollipop,' 'suck straw,' and 'gbe se enu' (put it in your mouth)
Slang term for oral sex that is used interchangeably to describe oral sex that is given by man/boy or woman/girl	'sucking,' 'sukky,' 'kiss it,' 'ice cream,' 'yoghurt,' and 'collabo'
Slang term for penile–anal sex	'back side,' 'bend down and sow,' 'backyard sex,' 'chopping from the back,' 'daudau' (penile thrusting from behind), 'doggy style,' 'gba eyin wole' (penetration from behind), 'hanging up,' 'load from the back,' 'monkey style,' 'oloshos' (sex worker's style), 'room,' 'redeemed,' 'sex through yansh,' 'shoki,' 'smash from the back,' 'table sex,' 'touching of toe,' 'wheel barrow' and 'yodi (project out your buttock or anus)'

Findings from the community & brothel surveys

1. Gender difference in oral sexual behaviours?

Pattern of oral sexual behaviours in men & women

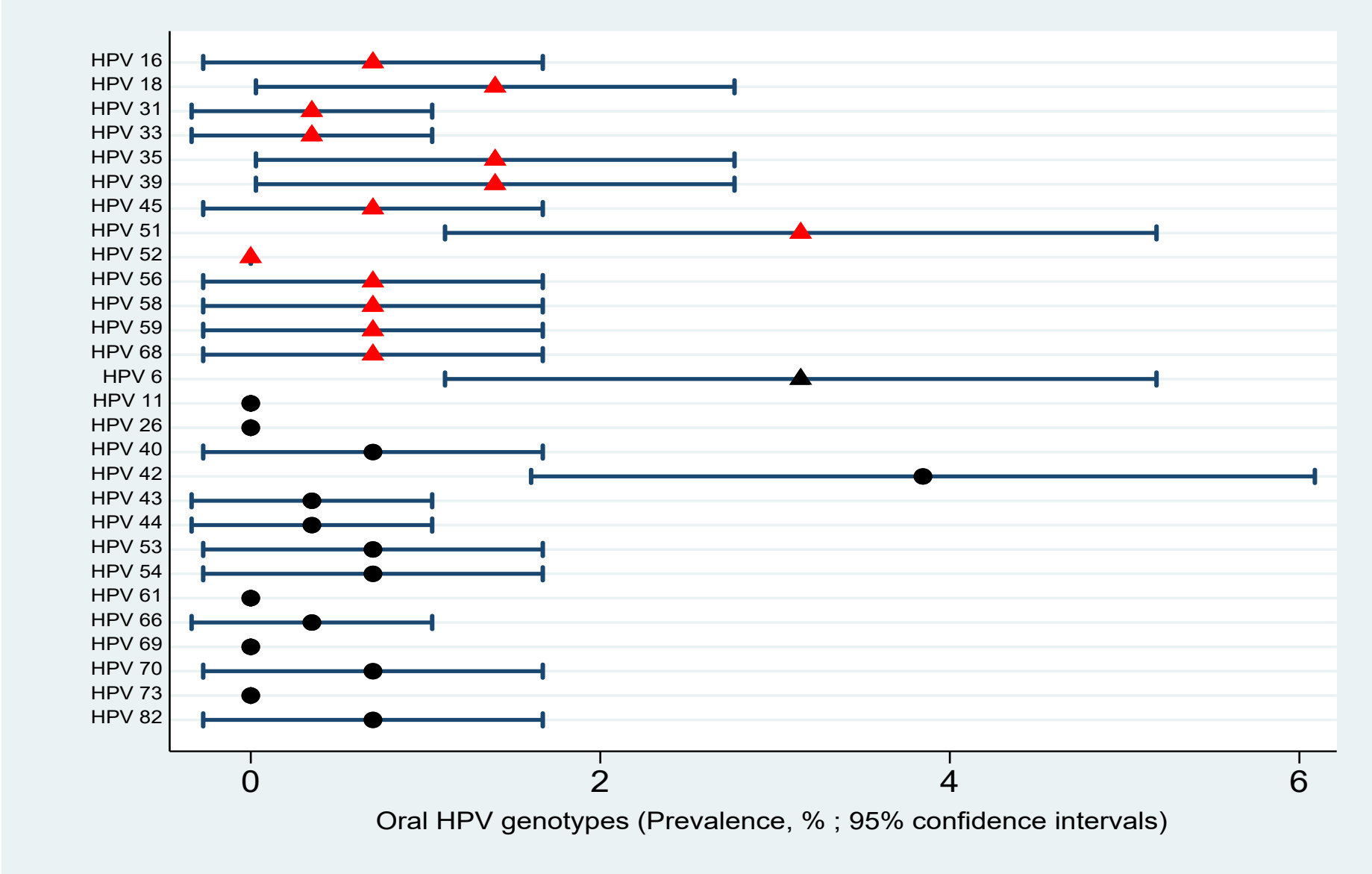


2. High prevalence of high-risk & low-risk HPV infections

Effect of age, HPV-35, risk factors

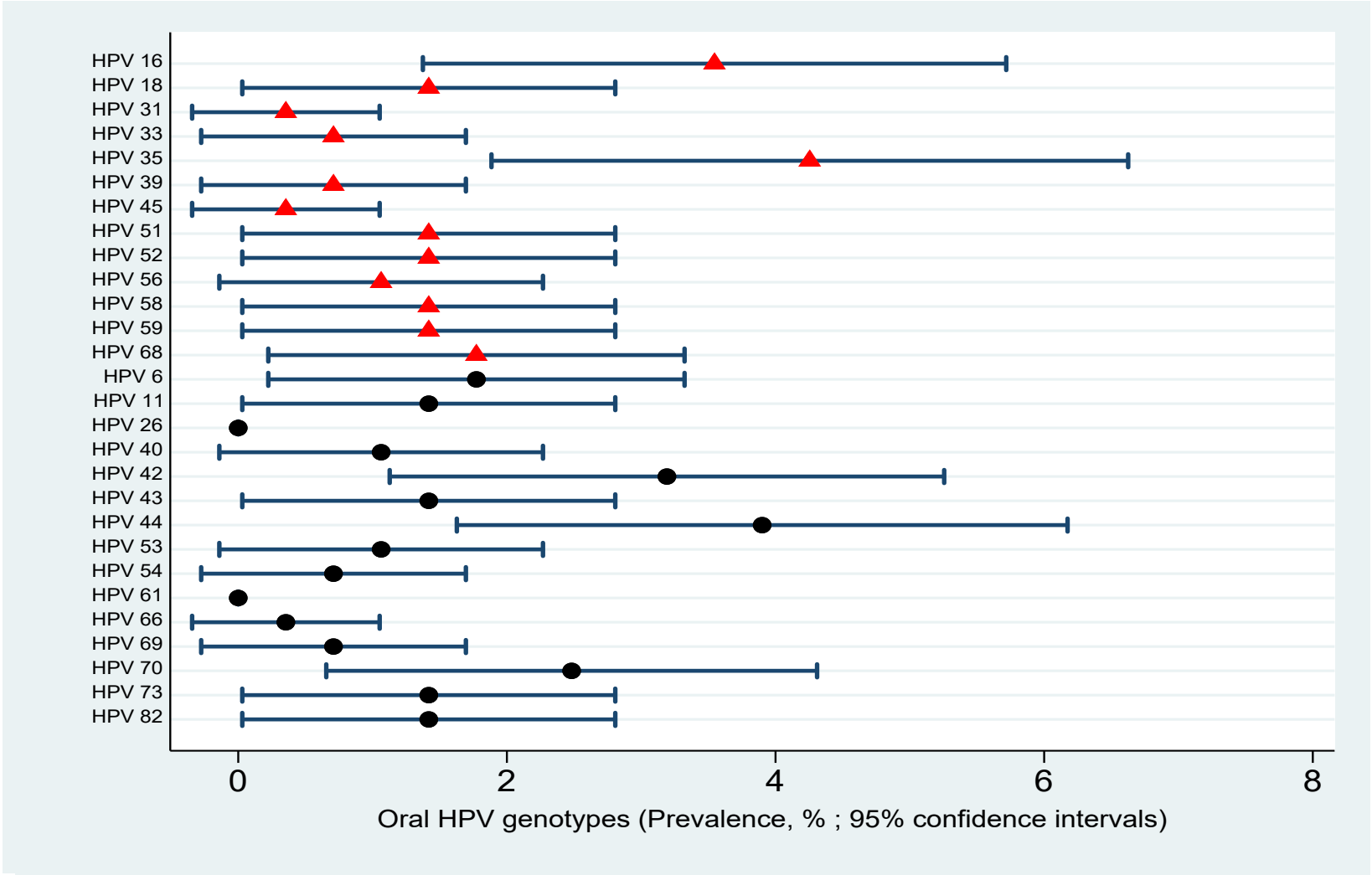
Prevalence of High-risk & Low-risk HPV by anatomic site in women (N=310)

RED triangle indicates HR-HPV point prevalence and BLACK BALL indicates LR-HPV point prevalence with lines indicating 95% CI



Prevalence of High-risk & Low-risk HPV by anatomic site in FSWs (N=315)

RED triangle indicates HR-HPV point prevalence and BLACK BALL indicates LR-HPV point prevalence with lines indicating 95% CI



Prevalence of Cervical, Vulvar, Anal and Oral HPV infections among women in Ibadan, Nigeria (N=310)

Variable	Cervical Sample		Vulvar Sample		Anal Sample		Oral Sample	
Genotype	n/N1	Prevalence % [95% CI]	n/N	Prevalence (%) [95% CI]	n/N	Prevalence (%) [95% CI]	n/N	Prevalence (%) [95% CI]
Any HPV		p=0.014*		p=0.523		p=0.021*		p=0.669
18-24 years	79/118	66.9 (57.8-75.3)	86/121	71.1 (62.1-79.0)	74/120	61.7 (52.4-70.4)	20/111	18.0 (11.4-26.4)
25-34 years	62/100	62.0 (51.7-71.5)	68/100	68.0 (57.9-77.0)	57/97	58.9 (48.3-68.7)	15/92	16.3 (9.4-25.5)
35-45 years	41/87	47.1 (36.3-58.1)	56/88	63.6 (52.7-73.6)	41/86	47.7 (36.8-58.7)	11/83	13.3 (6.8-22.5)
Overall	182/305	59.7 (53.9-65.2)	210/309	68.0 (62.4-73.1)	172/303	56.8 (51.0-62.4)	46/286	16.1 (12.0-20.9)
Any HR-HPV		p=0.095		p=0.142*		p=0.243		p=0.965
18-24 years	60/118	53.1 (44.7-61.3)	69/121	59.3 (51.0-67.3)	53/120	44.2 (35.1-53.5)	12/111	10.8 (5.7-18.1)
25-34 years	44/100	38.0 (28.1-48.8)	51/100	44.6 (34.2-55.3)	38/97	39.2 (29.4-49.6)	9/92	9.8 (4.6-17.8)
36-45 years	31/87	33.3 (22.2-46.0)	38/67	41.8 (29.8-54.5)	28/86	32.6 (22.8-43.5)	9/83	10.8 (5.1-19.6)
Overall	135/305	44.3 (38.6-50.0)	158/309	51.1 (45.4-56.8)	119/303	39.3 (33.7-45.0)	30/286	10.5 (7.2-14.6)
Any LR-HPV		p=0.001*		p=0.047*		p=0.495		p=0.891
18-24 years	60/118	50.8 (41.5-60.2)	66/121	54.5 (45.2-63.6)	56/120	46.7 (37.5-56.0)	11/111	9.9 (5.1-17.0)
25-34 years	41/100	41.0 (31.3-51.3)	50/100	50.0 (39.8-60.2)	44/97	45.4 (35.2-55.8)	8/92	8.7 (3.8-16.4)
35-45 years	22/87	25.3 (16.6-35.7)	33/88	37.5 (27.4-48.5)	33/86	38.4 (28.1-49.5)	9/83	10.8 (5.1-19.6)
Overall	123/305	40.3 (34.8-46.1)	149/309	48.2 (42.5-53.9)	133/303	43.9 (38.2-49.7)	28/286	9.8 (6.6-13.8)
Multiple HPV		p<0.001*		p=0.012*		p=0.014*		p=0.7215
18-24 years	54/118	45.8 (36.6-55.2)	59/121	48.7 (39.6-58.0)	51/120	42.5 (33.5-51.9)	4/111	3.6 (1.0-9.0)
25-34 years	29/100	29.0 (20.4-38.9)	42/100	42.0 (32.2-52.3)	31/97	32.0 (22.9-42.2)	4/92	4.3 (1.2-10.8)
35-45 years	18/87	20.7 (12.7-30.7)	25/88	28.4 (19.3-39.0)	20/86	23.3 (14.8-33.6)	5/83	6.0 (2.0-13.5)
Overall	101/305	33.1 (27.9-38.7)	126/309	40.8 (35.2-46.5)	102/303	33.7 (28.4-39.3)	13/286	4.5 (2.4-7.6)

Prevalence of cervical, vulvar, anal and oral HPV infections among female sex workers in Ibadan, Nigeria (N = 315)

Variable	Cervical Sample		Vulvar Sample		Anal Sample		Oral Sample	
Genotype	n/N	Prevalence (%) [95% CI]	n/N	Prevalence (%) [95% CI]	n/N	Prevalence (%) [95% CI]	n/N	Prevalence (%) [95% CI]
Any HPV		p=0.041*		p=0.001*		p=0.303		p=0.112
18-24 years	51/55	92.7 (82.4-98.0)	54/55	98.2 (90.3-99.9)	48/55	87.3 (75.5-94.7)	16/46	34.8 (21.4-50.2)
25-34 years	145/171	85.0 (78.5-89.8)	154/172	89.5 (84.0-93.7)	125/171	73.1 (65.8-79.6)	37/154	24.0 (17.5-31.6)
35-45years	67/87	77.0 (66.8-85.4)	69/88	78.4 (68.4-86.5)	59/85	69.4 (58.5-79.0)	15/82	18.3 (10.6-28.4)
Overall	263/313	84.0 (79.5-87.9)	277/315	87.9 (83.8-91.3)	232/311	74.6 (69.4-79.3)	68/282	24.1 (19.2-29.5)
Any HR-HPV		p=0.008		p<0.001*		p=0.006*		p=0.143
18-24 years	47/55	85.5 (73.3-93.5)	53/55	96.4 (87.5-99.6)	44/55	80.0 (67.0-89.6)	9/46	19.6 (9.4-33.9)
25-34 years	117/171	68.4 (60.9-75.3)	127/172	73.8 (66.6-80.2)	97/171	56.7 (48.9-64.3)	26/154	16.9 (11.3-23.8)
35-45years	53/87	60.9 (49.9-71.2)	57/88	64.8 (53.9-74.7)	48/85	56.5 (45.3-67.2)	7/82	8.5 (3.5-16.8)
Overall	217/313	69.3 (63.9-74.4)	237/315	75.2 (70.1-79.9)	189/311	60.8 (55.1-66.2)	42/282	14.9 (10.9-19.6)
Any LR-HPV		p=0.005*		p<0.001*		p=0.241		p=0.409
18-24 years	47/55	85.5 (73.3-93.5)	51/55	92.7 (82.4-98.0)	35/55	63.6 (49.6-76.2)	9/46	19.6 (9.4-33.9)
25-34 years	113/171	66.1 (58.5-73.1)	123/172	71.5 (64.1-78.1)	88/171	51.5 (43.7-59.2)	22/154	14.3 (9.2-20.8)
35-45years	52/87	59.8 (48.7-70.1)	51/88	58.0 (47.0-68.4)	43/85	50.6 (39.5-61.6)	9/82	11.0 (5.1-19.8)
Overall	212/313	67.7 (62.2-72.9)	225/315	71.4 (66.1-76.4)	166/311	53.4 (47.7-59.0)	40/282	14.2 (10.3-18.8)
Multiple HPV		p=0.001*		p<0.001*		p=0.001*		p=0.330
18-24 years	47/55	85.5 (73.3-93.5)	51/55	92.7 (82.4-98.0)	44/55	80.0 (67.0-89.6)	7/46	15.2 (6.3-28.9)
25-34 years	113/171	66.1 (58.5-73.1)	117/172	68.0 (60.5-74.9)	84/171	49.1 (41.4-56.9)	14/154	9.1 (5.1-14.8)
35-45years	48/87	55.2 (44.1-65.9)	53/88	60.2 (49.2-70.5)	44/85	55.1 (40.7-62.7)	6/82	7.3 (2.7-15.2)
Overall	208/313	66.5 (60.9-71.7)	221/315	70.2 (64.8-75.2)	172/311	55.3 (49.6-60.9)	27/282	9.6 (6.4-13.6)



Adjusted Model of Risk factors associated with cervical, vulvar, anal and oral HPV in women in Ibadan, Nigeria

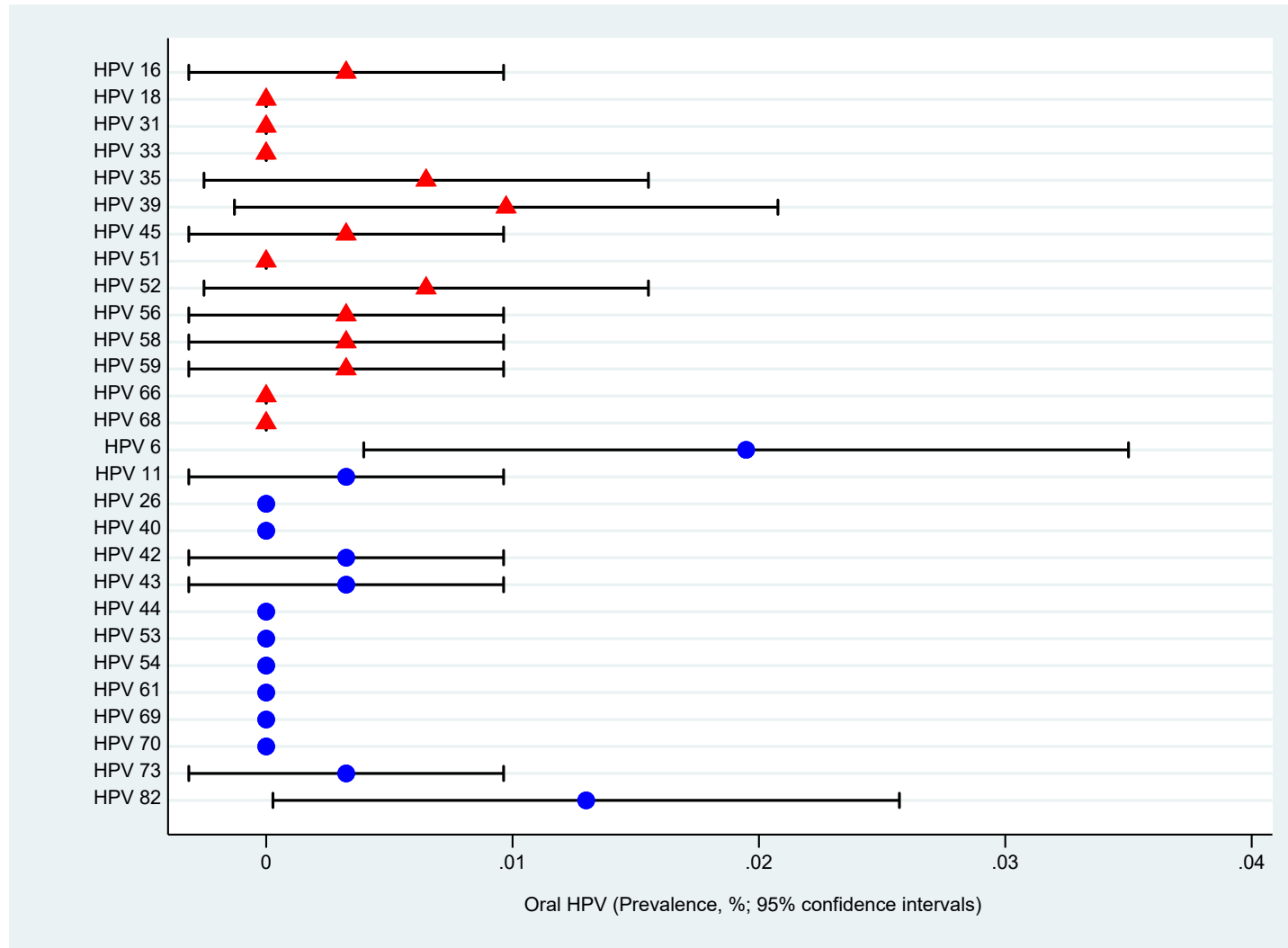
Variable	Cervical	Vulvar	Anal	Oral
	p-value Adjusted OR (95%CI)	p-value Adjusted OR (95%CI)	p-value Adjusted OR (95%CI)	p-value Adjusted OR (95%CI)
SOCIO-DEMOGRAPHIC FACTORS				
Study site	p =0.576	p=0.373	p=0.075	p=0.004
Mokola	1	1	1	1
Moniya/Sasa	0.82(0.51-1.31)	1.27(0.75-2.13)	0.63(0.38-1.05)	0.38(0.19-0.74)
Age group, years	p=0.018	p=0.509	p=0.521	p=0.693
18-24	1	1	1	1
25-34	0.81(0.46-1.41)	0.89(0.49-1.63)	0.96(0.54-1.72)	0.80(0.38-1.68)
35-45	0.44(0.25-0.78)	0.67(0.34-1.33)	0.70(0.37-1.35)	0.72(0.32-1.61)
Current marital status	p=0.334	p=0.027	p=0.073	p=0.497
Single	1	1	1	1
Married	0.79(0.43-1.46)	0.49(0.25-0.99)	0.47(0.24-0.91)	1.11(0.50-2.45)
Divorced/widowed/separated 5	1.67(0.48-5.73)	1.64(0.38-7.00)	0.44(0.13-1.48)	2.42(0.60-9.76)
BEHAVIOURAL FACTORS	Adjusted OR (95%CI)	Adjusted OR (95%CI)	Adjusted OR (95%CI)	Adjusted OR (95%CI)
Age difference between first vaginal sex partner and participant, years	p=0.133	p=0.026	p=0.050	p=0.570
≤ 5	1	1	1	1
≥ 6	0.78(0.46-1.31)	0.51(0.29-0.90)	0.58(0.34-1.00)	0.82(0.42-1.61)
number of vaginal sex partners (mean (SD))	p=0.002	p<0.001	p<0.001	p=0.404
	1.54(1.16-2.04)	2.03(1.38-3.01)	1.92(1.35-2.71)	1.09(0.89-1.33)
BIOLOGICAL FACTORS	Adjusted OR (95%CI)	Adjusted OR (95%CI)	Adjusted OR (95%CI)	Adjusted OR (95%CI)
Cervical HPV infection	NA	p<0.001	p<0.001	p=0.003
No		1	1	1
Yes		20.75(10.99-39.17)	4.10(1.85-9.11)	4.81(1.58-14.62)
Vulvar HPV infection	p<0.001	NA	p<0.001	p=0.371
No	1		1	1
Yes	12.85(5.70-28.99)		5.85(2.35-14.59)	0.57(0.17-1.95)
Anal HPV infection	p<0.001	p<0.001	NA	p=0.136
No	1	1		1
Yes	3.48(1.74-6.96)	15.31(8.12-28.86)		2.02(0.78-5.25)
Oral HPV infection	p<0.004	p=0.016	p=0.119	NA
No	1	1	1	
Yes	4.37(1.50-12.71)	2.53(1.13-5.67)	2.13(0.80-5.64)	



Adjusted Model of Risk factors associated with cervical, vulvar, anal and oral HPV infection among brothel-based FSWs in Ibadan, Nigeria.

Variable	Cervical	Vulvar	Anal	Oral
	p-value Adjusted OR (95%CI)	p-value Adjusted OR (95%CI)	p-value Adjusted OR (95%CI)	p-value Adjusted OR (95%CI)
SOCIO-DEMOGRAPHIC FACTORS				
Age group, years	p=0.035	p<0.001	p=0.052	p=0.120
18-24	1	1	1	1
25-34	0.44 (0.15-1.31)	0.16 (0.02-1.22)	0.39 (0.16-0.93)	0.59 (0.29-1.21)
35-45	0.26 (0.08-0.82)	0.07 (0.01-0.52)	0.37 (0.14-0.93)	0.42 (0.18-0.96)
Income per month	P=0.657	p=0.819	p=0.030	p=0.515
≤20,000N (≤56 USD)	1	1	1	1
20,001 - 40,000N (>56 – 112USD)	0.94 (0.43-2.02)	1.24 (0.53-2.90)	2.07 (1.11-3.86)	0.69 (0.35-1.36)
> 40,000N (>112USD)	0.71 (0.33-1.55)	0.96 (0.41-2.28)	2.10 (1.09-4.04)	0.73 (0.37-1.44)
BEHAVIOURAL FACTORS	Adjusted OR (95%CI)	Adjusted OR (95%CI)	Adjusted OR (95%CI)	Adjusted OR (95%CI)
Female genital mutilation	p=0.019	p=0.777	p=0.327	p=0.452
No	1	1	1	1
Yes	0.48 (0.26-0.89)	1.11 (0.54-2.26)	0.77 (0.45-1.30)	0.80 (0.45-1.42)
Ever heard of HPV	p=0.283	p=0.165	p=0.147	p=0.029
No	1	1	1	1
Yes	0.49 (0.14-1.69)	0.39 (0.11-1.36)	0.42 (0.13-1.33)	3.88 (1.19-12.64)
BIOLOGICAL FACTORS	Adjusted OR (95%CI)	Adjusted OR (95%CI)	Adjusted OR (95%CI)	Adjusted OR (95%CI)
RDT HIV final test result	p=0.014	p=0.545	p=0.644	p=0.027
Negative	1	1	1	1
Positive	11.39 (1.03-125.83)	1.66 (0.30-9.09)	1.27 (0.80-4.84)	2.40 (1.12-5.14)
Cervical HPV infection	NA	p<0.001	p=0.151	p=0.307
No		1	1	1
Yes		6.48 (2.70-15.57)	1.97 (0.80-4.84)	1.75 (0.57-5.29)
Vulvar HPV detection	p<0.001	NA	p<0.001	p=0.503
No	1		1	1
Yes	16.60 (5.80-47.54)		10.55 (3.67-30.31)	1.63 (0.37-7.11)
Anal HPV detection	p=0.281	p<0.001	NA	p=0.349
No	1	1		1
Yes	1.65 (0.69-3.99)	8.88 (3.66-23.28)		1.47(0.65-3.34)

Prevalence of High-risk & Low-risk HPV by anatomic site in men (N=310)



Prevalence of Penile, Anal and Oral HPV in men in Ibadan, Nigeria (N=316)

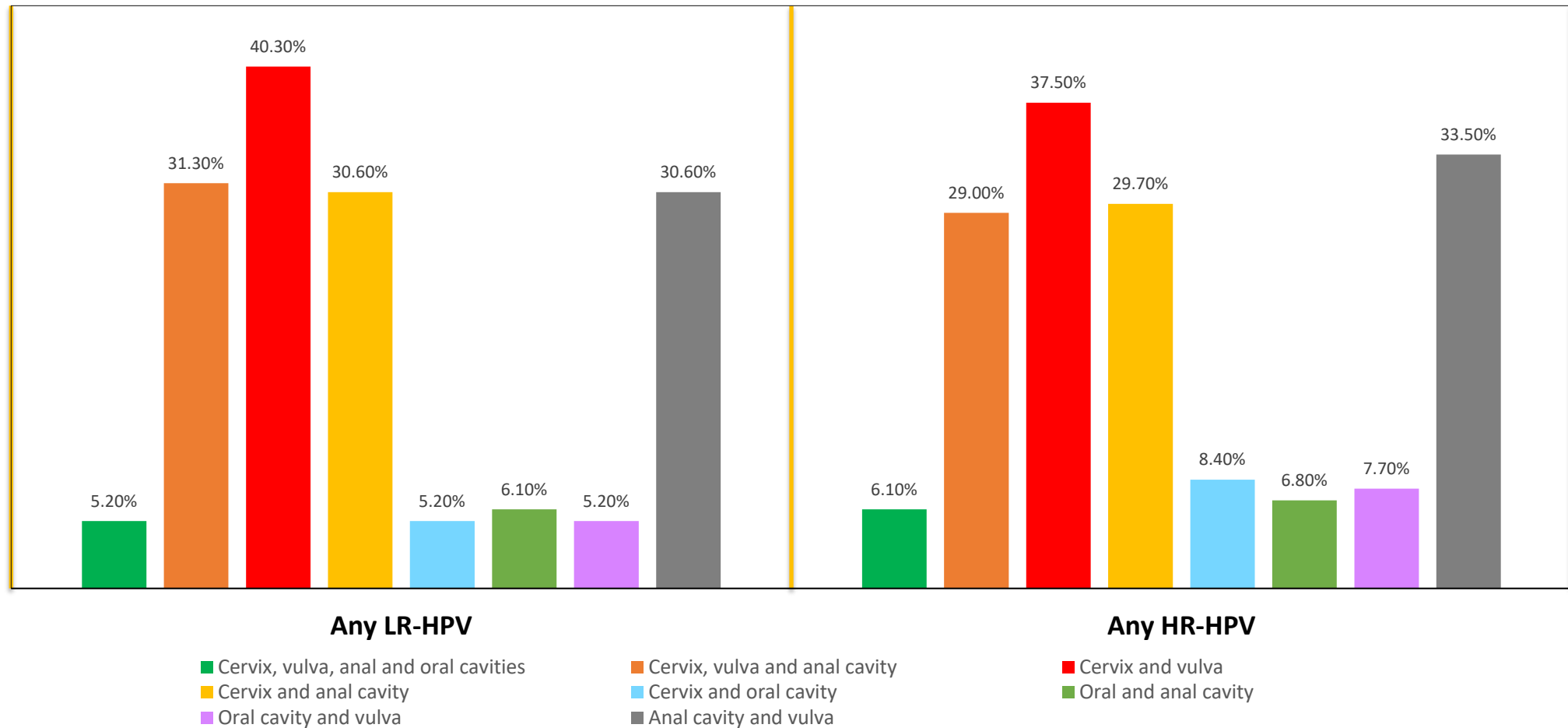
Variable	Penile Sample		Anal Sample		Oral Sample	
Genotype	n/N	Prevalence % [95% CI]	n/N	Prevalence (%) [95% CI]	n/N	Prevalence (%) [95% CI]
Any HPV		p=0.201		p=0.987		p=0.012*
18-24 years	34/137	24.8 (17.8-32.9)	10/105	9.5 (4.7-16.8)	8/135	5.9 (2.6-11.3)
25-34 years	50/94	53.2 (42.6-63.6)	7/72	9.7 (4.0-19.0)	7/93	7.5 (3.0-14.9)
35-45 years	42/80	52.5 (41.0-63.8)	6/61	9.8 (3.7-20.2)	9/80	11.3 (5.3-20.3)
Overall	126/311	40.5 (35.0-46.2)	23/238	9.7 (6.2-14.1)	24/308	7.8 (5.1-11.4)
Any HR-HPV		p=0.003*		p=0.564		p=0.028*
18-24 years	23/137	16.8 (11.0-24.1)	23/137	16.8 (11.0-24.1)	4/135	3.0 (0.8-7.4)
25-34 years	42/94	44.7 (34.4-55.3)	42/94	44.7 (34.4-55.3)	3/93	3.2 (0.7-9.1)
36-45 years	31/80	38.8 (28.1-50.3)	31/80	38.8 (28.1-50.3)	4/80	5.0 (1.4-12.3)
Overall	96/311	30.8 (25.7-36.2)	96/312	30.8 (25.7-36.2)	11/308	3.6 (1.8-6.3)
Any LR-HPV		p=0.007*		p=0.041*		p<0.001*
18-24 years	20/137	14.6 (9.2-21.6)	5/105	4.8 (1.6-10.8)	4/135	3.0 (0.8-7.4)
25-34 years	22/94	23.4 (15.3-33.3)	2/72	2.8 (0.3-9.7)	4/93	4.3 (1.2-10.6)
35-45 years	28/80	35.0 (24.7-46.5)	3/61	4.9 (1.0-13.7)	6/80	7.5 (2.8-15.6)
Overall	70/311	22.4 (17.9-27.5)	10/238	4.2 (2.0-7.6)	14/308	4.5 (2.5-7.5)
Multiple HPV		p<0.001*		p=0.771		p=0.488
18-24 years	11/137	8.0 (4.1-13.9)	3/105	2.9 (0.6-8.1)	-	-
25-34 years	25/94	26.6 (18.0-36.7)	2/72	2.8 (0.3-9.7)	1/93	1.1 (0.0-5.8)
35-45 years	23/80	28.8 (19.2-40.0)	2/61	3.3 (0.4-11.3)	1/80	1.3 (0.0-6.8)
Overall	59/311	19.0 (14.8-23.8)	7/238	2.9 (1.2-6.0)	2/308	0.6 (0.0-2.3)

VARIABLE	PENILE SITE HPV	ANAL SITE HPV	ORAL SITE HPV
	p-value Adjusted OR (95%CI)	p-value Adjusted OR (95%CI)	p-value Adjusted OR (95%CI)
SOCIO-DEMOGRAPHIC (Level 1)			
Age group	p=0.006	p=0.663	p=0.967
18-24 years	1	1	1
25-34 years	3.05 (1.52-6.13)	0.66 (0.23-1.90)	1.21 (0.14-10.23)
35-45 years	2.67 (1.14-6.26)		1.33 (0.15-11.93)
Current marital status	p=0.675	p=0.595	p<0.001*
Living with partner	1	1	1
Not living with partner	0.87 (0.44-1.70)	0.78 (0.31-1.94)	0.03 (0.00-0.38)
Have other sexual partner	p=0.115	p=0.907	p=0.002*
No	1	1	1
Yes	1.95 (0.85-4.51)	1.08 (0.32-3.59)	39.61 (3.71-423.18)
BEHAVIOURAL FACTORS (Level 2)			
Number of lifetime partners from which oral sex was received	p=0.034 0.62 (0.39-1.01)	NA	NA
BIOLOGICAL FACTORS (Level 3)			
Penile HPV Infection		p<0.001	p=0.138
No	NA	1	1
Yes		15.36 (3.40-69.45)	2.53 (0.71-9.00)
Anal HPV Infection	p<0.001		p=0.283
No	1	NA	1
Yes	21.32 (4.38-103.86)		2.60 (0.49-13.71)

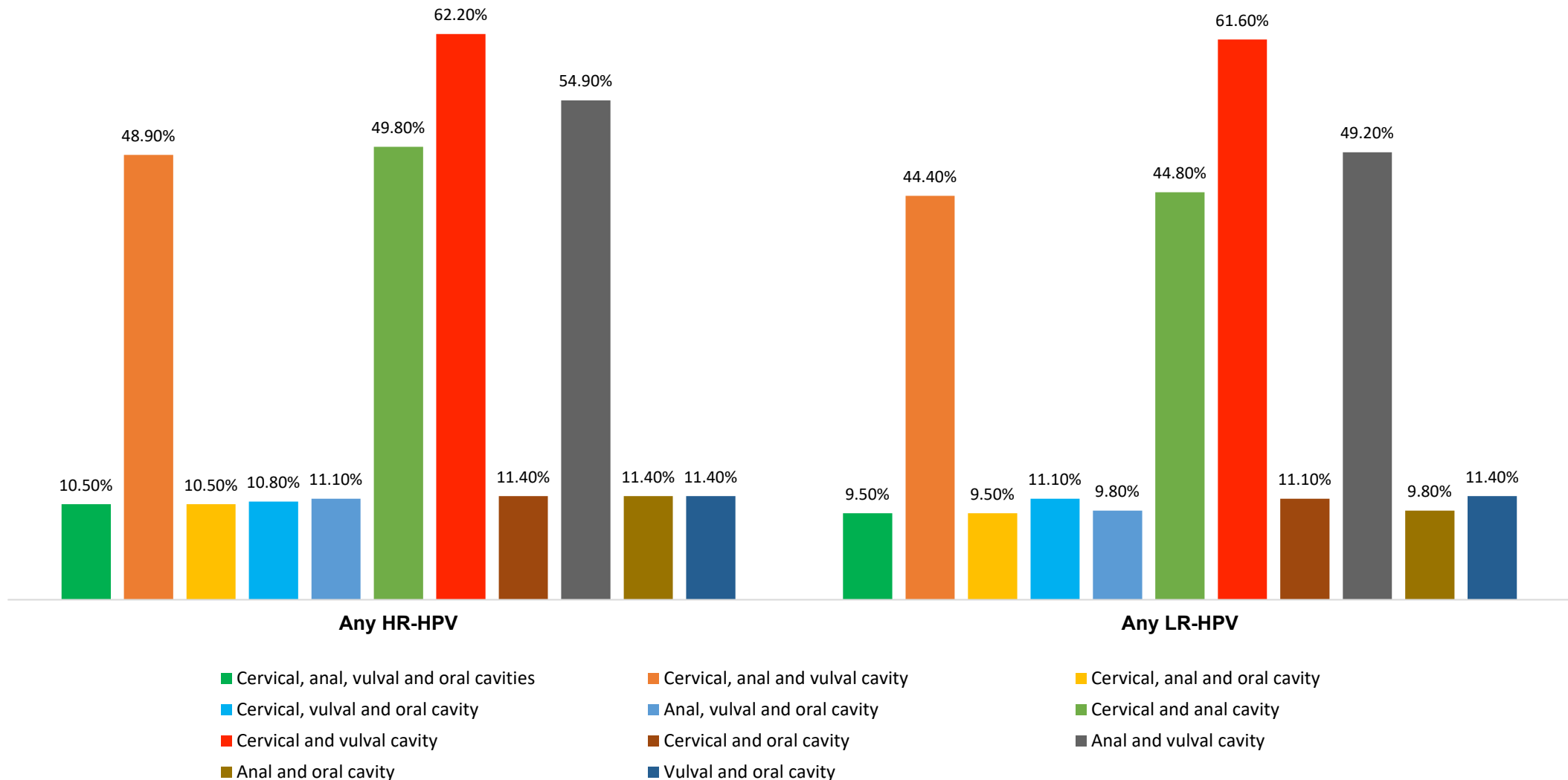
3. Concordance of genotype specific HPV infections

Viral shedding, risk of synchronous HPV associated cancers & public health implication for screening

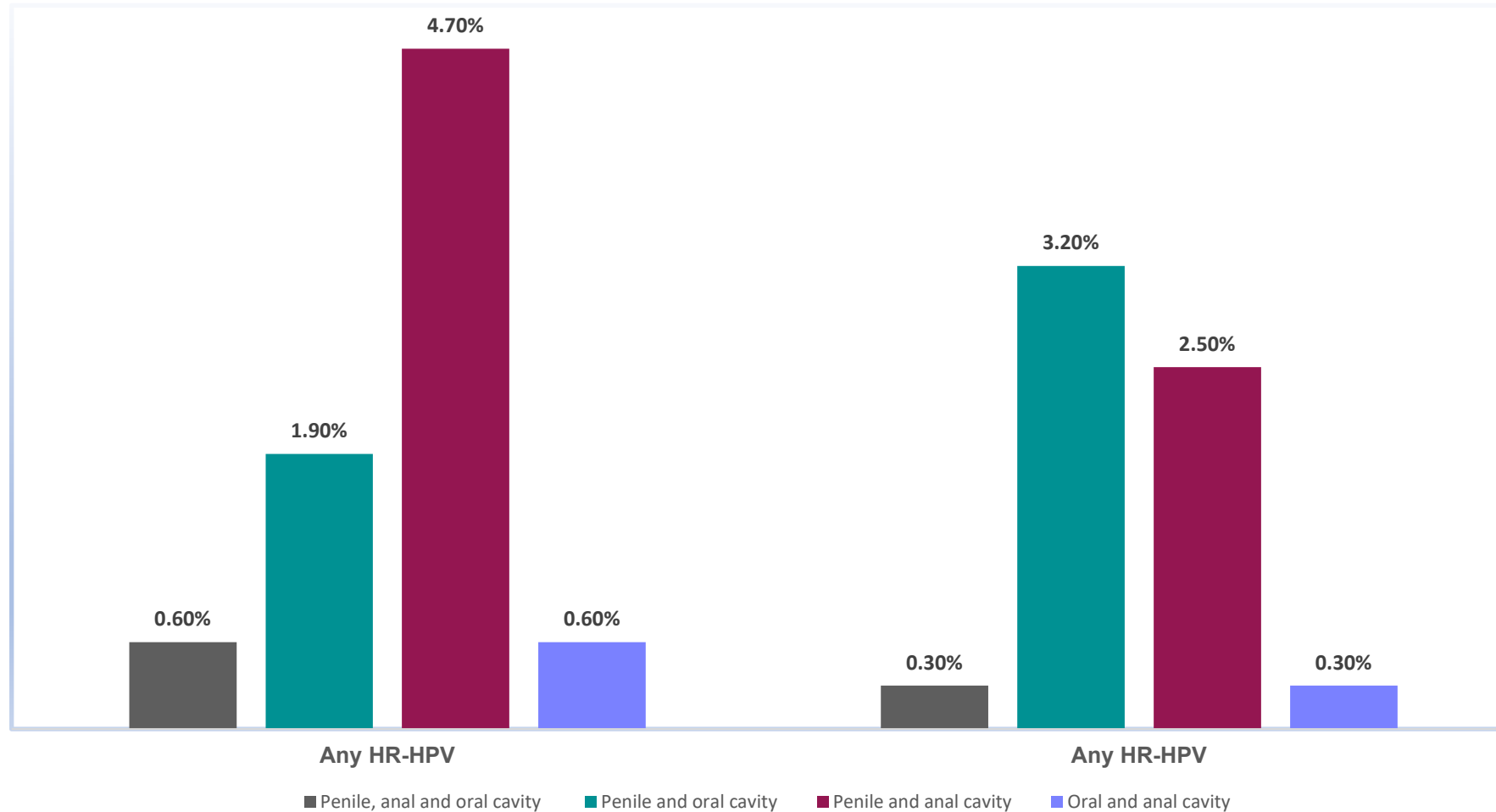
Concordance of HPV specific genotypes by anatomical sites among women in Ibadan, Nigeria (n=310)



Concordance of HPV specific genotypes by anatomical sites among FSWs in Ibadan, Nigeria (n=315)



Concordance of HPV specific genotypes by anatomical sites among men in Ibadan, Nigeria (n=316)





Public health message from SHINI study

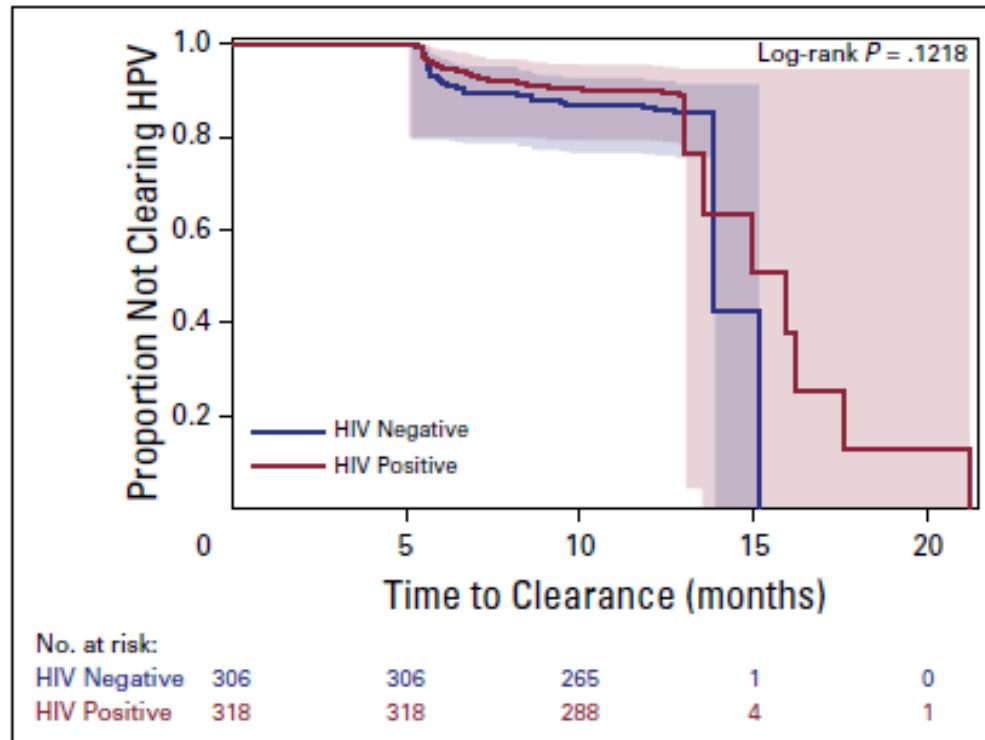
- Adolescents and young adults in are engaging in wide range of penetrative and non-penetrative sexual risk behaviours
 - May suggest increasing risk of oral and anal STIs including precancer & cancers
- The study provides a robust information on the epidemiology of HPV infections in genital sites – cervix, vulvar and penile; anal and oral sites in women (including FSWs) and in Nigeria
 - HPV 35 not covered in any candidate HPV vaccine is increasingly reported in SSA
 - May suggest increasing risk of oral and anal HPV associated cancers
- Concordance of HPV infections has programmatic and public health implication in the design of control for HPV associated cancers in women



HIV and HPV Infections/HACs

HIV and burden of high-risk HPV in the cervix

Adebamowo et al 2013



Ezeachi et al 2014

multivariate logistic analysis

Variables at enrolment	Initial model ^a OR (95% CI)	2nd Model ^b OR (95% CI)	Final model ^c OR (95% CI)
HIV status (515)			
Negative	1 (ref)	1 (ref)	1 (ref)
Positive	1.7 (1.3 -2.5)	1.7 (1.3-2.5)	1.8 (1.4-2.2)
Antiretroviral drug use (220)			
Not on drugs	1 (ref)	1 (ref)	1 (ref)
On drugs	0.4 (0.3 - 0.9)	0.4 (0.3 - 0.9)	0.4 (0.3-0.5)
CD4 cell count (220)			
<200	2.9 (1.3-7.1)	2.5 (1.5-6.1)	2.4 (1.7-5.9)
200 – 499	1 (ref)	1 (ref)	1 (ref)
500 and above	0.6 (0.3-1.0)	0.6 (0.5-0.9)	0.7 (0.5-0.8)
Viral load (220)			
<1000	0.5 (0.2-1.9)	0.6 (0.4 -1.3)	0.6 (0.3 -1.4)
1000 – 10,000	1 (ref)	1 (ref)	1 (ref)
≥ 10,000	3.1 (1.1-9.9)	3.1 (0.9- 8.7)	2.7 (0.6- 8.1)

^aadjusted for age ^badjust for HIV status and antiretroviral drug use ^cin addition to a and c adjusted for type of community, life time sexual partner and marital status.

RESEARCH ARTICLE

Open Access

High-risk human papilloma virus and cervical abnormalities in HIV-infected women with normal cervical cytology

Jonah Musa^{1,6*}, Chad Achenbach², Babafemi Taiwo², Baiba Berzins², Olugbenga Silas³, Patrick H Daru¹, Oche Agbaji^{4,6}, Godwin Imade^{1,6}, Atiene S Sagay^{1,6}, John A Idoko⁴, Phyllis J Kanki⁵ and Robert L Murphy²

Variable	Odds ratio (OR)	95% CI	P-Value
CD4 < 350 cells/mm ³	1.03	0.30-3.48	0.968
Positive HR-HPV (HC2)	4.00	1.08-14.75	0.037
Previous abortions	0.73	0.21-2.51	0.617
Age ≥30 years	3.58	0.73-17.45	0.115
Viral load >400 RNA copies/ml	0.45	0.10-2.22	0.33
Positive HR-HPV (HC2)	5.44 (AOR)	1.37-21.50	0.016
Age ≥30 years	4.87 (AOR)	0.91-26.17	0.065

AOR = Adjusted Odds Ratio of having an abnormal colposcopic examination findings in a model of positive HR-HPV and Age ≥30 years as co-variates (Hosmer-Lemeshow goodness-of-fit p-value 0.782).

Outcome	Positive HR-HPV	Negative HR-HPV	P-value
Mean follow up time (Months)	20.5 ± 4.1	19.9 ± 4.3	0.554*
Colposcopic finding			
Abnormal	9(29.0)	22(71.0)	0.001 [†]
Normal	3(6.4)	44(93.6)	
Cervical biopsy taken			
Yes	10(83.3)	2(16.7)	0.001 [†]
No	21(32.8)	43(67.2)	
Histologic diagnosis			
Normal cervix	1(33.3)	2(66.7)	
CIN I	3(75.0)	1(25.0)	
CIN II	0(0.0)	1(100.0)	
CIN III	2(100.0)	0(0.0)	
Cervical Carcinoma in situ (CIS)	2(100.0)	0(0.0)	
Proportion of CIN 2+ detected	0.133	0.021	0.0494 [‡]

*Student t-test for means, [†]Fisher's exact p-values, [‡]t-test of proportion.

Human papillomavirus prevalence and type distribution in invasive cervical cancer in sub-Saharan Africa

Lynette Denny¹, Isaac Adewole², Rose Anorlu³, Greta Dreyer⁴, Manivasan Moodley⁵, Trudy Smith⁶, Leon Snyman⁴, Edwin Wiredu⁷, Anco Molijn⁸, Wim Quint⁸, Gunasekaran Ramakrishnan⁹ and Johannes Schmidt¹⁰

HPV 16, 18 & 35 commonest hr-HPV types by country & histologic types

HPV type ¹	Ghana (N ² = 121)	Nigeria (N ² = 145)	South Africa (N ² = 181)	Total (N ² = 447)
HPV16	55.4 [46.1, 64.4]	52.4 [44.0, 60.8]	47.5 [40.1, 55.1]	51.2 [46.5, 56.0]
HPV18	19.0 [12.4, 27.1]	14.5 [9.2, 21.3]	18.2 [12.9, 24.6]	17.2 [13.8, 21.1]
HPV35	6.6 [2.9, 12.6]	9.0 [4.9, 14.8]	9.9 [6.0, 15.3]	8.7 [6.3, 11.7]
HPV45	6.6 [2.9, 12.6]	7.6 [3.8, 13.2]	7.7 [4.3, 12.6]	7.4 [5.1, 10.2]
HPV33	3.3 [0.9, 8.2]	1.4 [0.2, 4.9]	6.6 [3.5, 11.3]	4.0 [2.4, 6.3]
HPV52	2.5 [0.5, 7.1]	2.8 [0.8, 6.9]	1.7 [0.3, 4.8]	2.2 [1.1, 4.1]
HPV51	1.7 [0.2, 5.8]	2.8 [0.8, 6.9]	1.1 [0.1, 3.9]	1.8 [0.8, 3.5]
HPV31	0.8 [0.0, 4.5]	2.1 [0.4, 5.9]	1.7 [0.3, 4.8]	1.6 [0.6, 3.2]
HPV68/73	0.8 [0.0, 4.5]	0.0 [0.0, 2.5]	2.8 [0.9, 6.3]	1.3 [0.5, 2.9]
HPV56	1.7 [0.2, 5.8]	0.7 [0.0, 3.8]	1.1 [0.1, 3.9]	1.1 [0.4, 2.6]
HPV59	0.0 [0.0, 3.0]	2.8 [0.8, 6.9]	0.6 [0.0, 3.0]	1.1 [0.4, 2.6]
HPV39	0.8 [0.0, 4.5]	1.4 [0.2, 4.9]	0.6 [0.0, 3.0]	0.9 [0.2, 2.3]
HPV58	0.8 [0.0, 4.5]	1.4 [0.2, 4.9]	0.0 [0.0, 2.0]	0.7 [0.1, 1.9]
HPV53	0.0 [0.0, 3.0]	0.7 [0.0, 3.8]	0.0 [0.0, 2.0]	0.2 [0.0, 1.2]
HPV66	0.0 [0.0, 3.0]	0.0 [0.0, 2.5]	0.6 [0.0, 3.0]	0.2 [0.0, 1.2]
HPV70	0.0 [0.0, 3.0]	0.7 [0.0, 3.8]	0.0 [0.0, 2.0]	0.2 [0.0, 1.2]

¹In 11 HPV positive women, the HPV type could not be determined.

²N = total number of women infected with a single HPV type.

HPV type	Single ICC ¹ (N ² = 447)	SCC ³ (N ² = 394)	ADC ⁴ (N ² = 27)
HPV16	51.2 [46.5, 56.0]	53.8 [48.7, 58.8]	22.2 [8.6, 42.3]
HPV18	17.2 [13.8, 21.1]	15.0 [11.6, 18.9]	29.6 [13.8, 50.2]
HPV35	8.7 [6.3, 11.7]	9.4 [6.7, 12.7]	3.7 [0.1, 19.0]
HPV45	7.4 [5.1, 10.2]	7.1 [4.8, 10.1]	18.5 [6.3, 38.1]
HPV33	4.0 [2.4, 6.3]	3.8 [2.1, 6.2]	3.7 [0.1, 19.0]
HPV52	2.2 [1.1, 4.1]	2.3 [1.0, 4.3]	–
HPV51	1.8 [0.8, 3.5]	2.0 [0.9, 4.0]	–
HPV31	1.6 [0.6, 3.2]	1.5 [0.6, 3.3]	3.7 [0.1, 19.0]
HPV68/ 73	1.3 [0.5, 2.9]	1.3 [0.4, 2.9]	3.7 [0.1, 19.0]
HPV56	1.1 [0.4, 2.6]	0.8 [0.2, 2.2]	7.4 [0.9, 24.3]
HPV59	1.1 [0.4, 2.6]	0.8 [0.2, 2.2]	7.4 [0.9, 24.3]
HPV39	0.9 [0.2, 2.3]	1.0 [0.3, 2.6]	–
HPV58	0.7 [0.1, 1.9]	0.5 [0.1, 1.8]	–
HPV53	0.2 [0.0, 1.2]	0.3 [0.0, 1.4]	–
HPV66	0.2 [0.0, 1.2]	0.3 [0.0, 1.4]	–
HPV70	0.2 [0.0, 1.2]	0.3 [0.0, 1.4]	–



Drivers of HPV Associated Cancers in SSA....1

- **Increased prevalence and or incidence of sexual risk behaviors**
 - Poor awareness/Knowledge
 - Misconceptions/Myths
 - Socio-cultural interpretations
 - Lifestyle
- **Health seeking behaviors**
 - Interpretation of a healthy person
 - Concept of disease risk
 - Influence of culture and belief system
- **Poor understanding of connection between sexual behaviors & health risks**



Drivers of HPV Associated Cancers in SSA...2

- **Insufficient or complete of lack of data to understand the burden, biological mechanism, risk profiling, diagnosis & treatment**
 - Most studies are not well designed to measure specific outcome, generalisability & challenge of social desirability effect
 - Lack of national data on non-penile vaginal sex to appreciate the impact of different sexual behaviors
 - Extremely low number of studies to understand basic science and biological mechanism of HACs in SSA
- **Weak health system**
 - Capacity in human resource, opportunity and equipment – collapse of basic health facilities
 - Low investment – vaccination, screening & treatment opportunity - all are largely by out of pocket - >70% across many countries in SSA
 - Inefficient referral system

HPV Vaccination access is disproportionately lower in countries in SSA

L. Bruni et al.

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High-income countries



Low- and middle- income countries

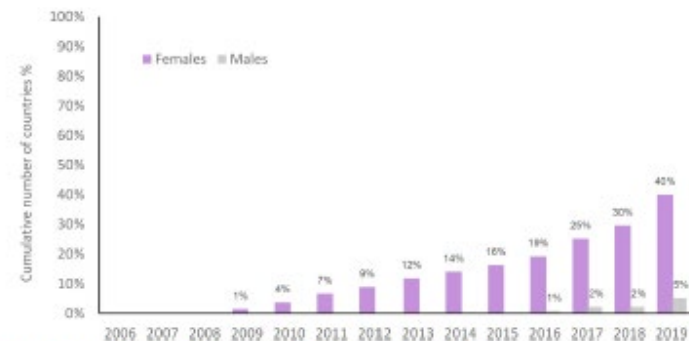
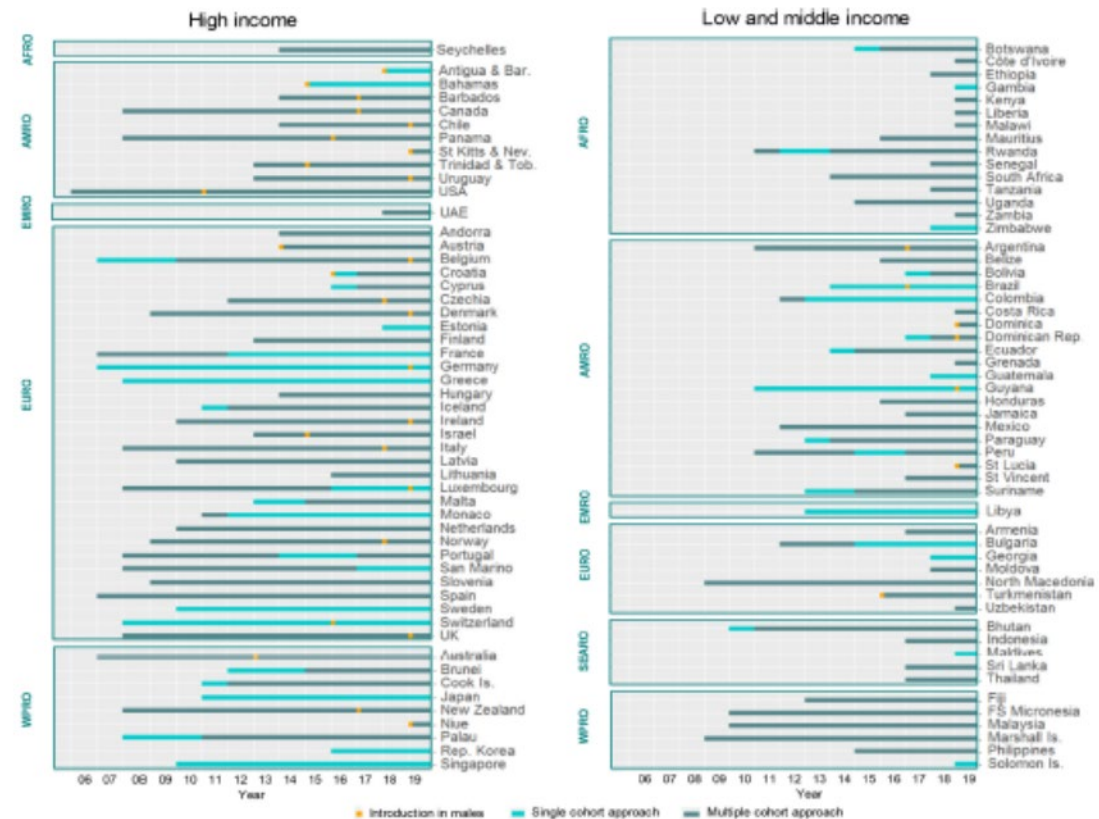


Fig. 2. Cumulative number of countries that have introduced HPV vaccination by sex, year and income level. Romania, Lesotho, Kazakhstan stopped their programs in 2011, 2012, and 2015, respectively. In Japan although the vaccine remains in the national immunization program and is free for 12–16-year-old girls, all proactive recommendations for it are suspended since 2012. Peru also temporarily stopped the program between 2012 and 2013.

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HPV Vaccination access is disproportionately lower in countries in SSA....2 — (The Gambia, Liberia, Cote d'Ivoire, Kenya, Malawi and Zambia recently introduced HPV vaccination)

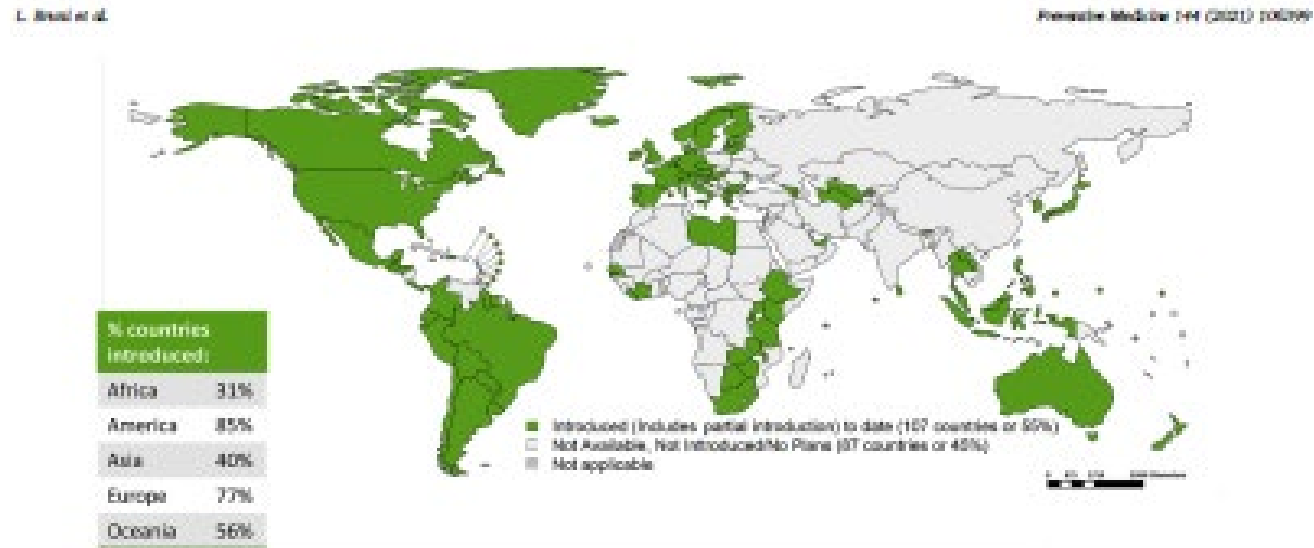


Fig. 1. WHO member states with HPV vaccination in their national immunization program, as of June 2020. It does not include territories, state of free-association, or semi-autonomous regions. The boundaries and names shown and the designations used on this map do not imply the expression of any opinion whatsoever concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

Recommendations

- **Awareness creations on different sexual behaviors and concept of safe sexual practices**
 - Family life education program/Sexual Health Education
 - **Strengthening of sexual health policy and programs** – many countries in SSA need to review & invest in Sexual Health Programs
- **National Data on different Sexual Behaviors in long overdue**
 - NATSAL in UK & NHANES in US
 - HIV & HPV can be nested within the national surveys
- **Robust Cancer Control Plan that will include HACs**
 - BCC, Vaccination & Screening – single dose vaccination & Point of care hrHPV testing triage & treat may be a game changer in SSA
 - Self sampling for HPV tests

Recommendations

- **Sustainable access to cancer prevention & treatment**
 - Health insurance vs Out-of-pocket
 - Investment in capacity building – Training/skill acquisition
 - Partnerships that works with established centers – within and outside SSA
- **Generate evidence to answer contextual challenges and proffer solutions to HACs**
 - Cutting edge research to answer critical questions – Why is HPV-35 have relative predilection for black people including African-American?
- **Data biorepository**
 - Sexual behaviors
 - Biological samples
 - Population Cancer Registry – Very limited

Conclusion

- There is limited information on the relationship between different sexual behaviors and risk of HPV infections and associated cancers
- Limited studies on Non-penile vaginal sex poorly defined their outcome/did not provide definition making it difficult to understand the impact of the risk
- Collaborative research at the population level will be valuable in understanding the role of sexual behaviors in HPV associated cancers across different populations

Current collaborations

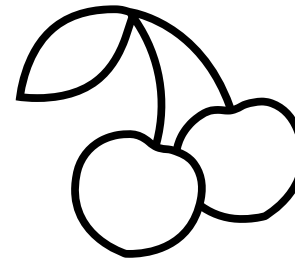
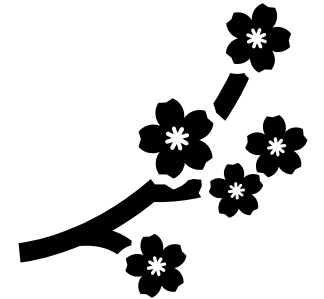
1. The Natural history of oral HPV infections, precancer & cancer in men and women involved in heterosexual relationship is poorly understood – **R01 2022- Epigenetics of HPV associated oral and oropharyngeal cancers in PLWH**
2. The high prevalence risk of high-risk HPV infection in multiple anatomic sites need to be further investigated – Synchronous risk of cancer? – **Epigenetic of anal HPV in women with cervical high-risk HPV - BIGCAT Award 2022**
3. Access to screening methods - **Implementation science on Self sampling for cervical cancer screening in WA – U01 Award 2022**
4. What is the pattern of non-penile vaginal sexual behaviours in people living with HIV before and after their diagnosis? How does these behaviours influence their risk to HPV associated cancers?

Acknowledgements

- College of Medicine, University of Ibadan, Ibadan, Nigeria
- London School of Hygiene and Tropical Medicine, London
- Collaborators & Mentors
- Oyo State Government
- Study team members
- Leadership of study sites & participants



Thank you for
Listening!



Introduction

- Human papillomavirus (HPV) associated cancers are increasingly reported globally including sub-Saharan Africa (SSA); it currently accounts for 4.5% of all cancers with highest burden in low-and middle- income countries
- The persistence of HPV infection is associated with anogenital warts and in over 95.0% of cervical cancer, 88.0% of anal cancer, 70.0% of vaginal cancers, 43.0% of vulvar cancer and 25.6% of head and neck cancers
- However, data on the epidemiology of oral and other genital HPV infections with the exception of cervical HPV infection are limited in SSA

RESEARCH ARTICLE

Oral, genital and anal human papillomavirus infections among female sex workers in Ibadan, Nigeria

Imran O. Morhason-Bello^{1,2,3*}, Kathy Baisley⁴, Miquel A. Pavon^{5,6}, Isaac F. Adewole¹, Rasheed A. Bakare⁷, Silvia de Sanjosé⁸, Suzanna C. Francis⁹, Deborah Watson-Jones^{3,10}

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RESEARCH ARTICLE

Open Access



Prevalence and genotype specific concordance of oro-genital and anal human papillomavirus infections among sexually active Nigerian women

Imran O. Morhason-Bello^{1,2*} , Kathy Baisley³, Miquel Angel Pavon⁴, Isaac F. Adewole¹, Rasheed Bakare⁵, Silvia de Sanjosé^{4,6}, Suzanna C. Francis³ and Deborah Watson-Jones^{7,8}



Oral sex leather

Types of HPV Vaccines

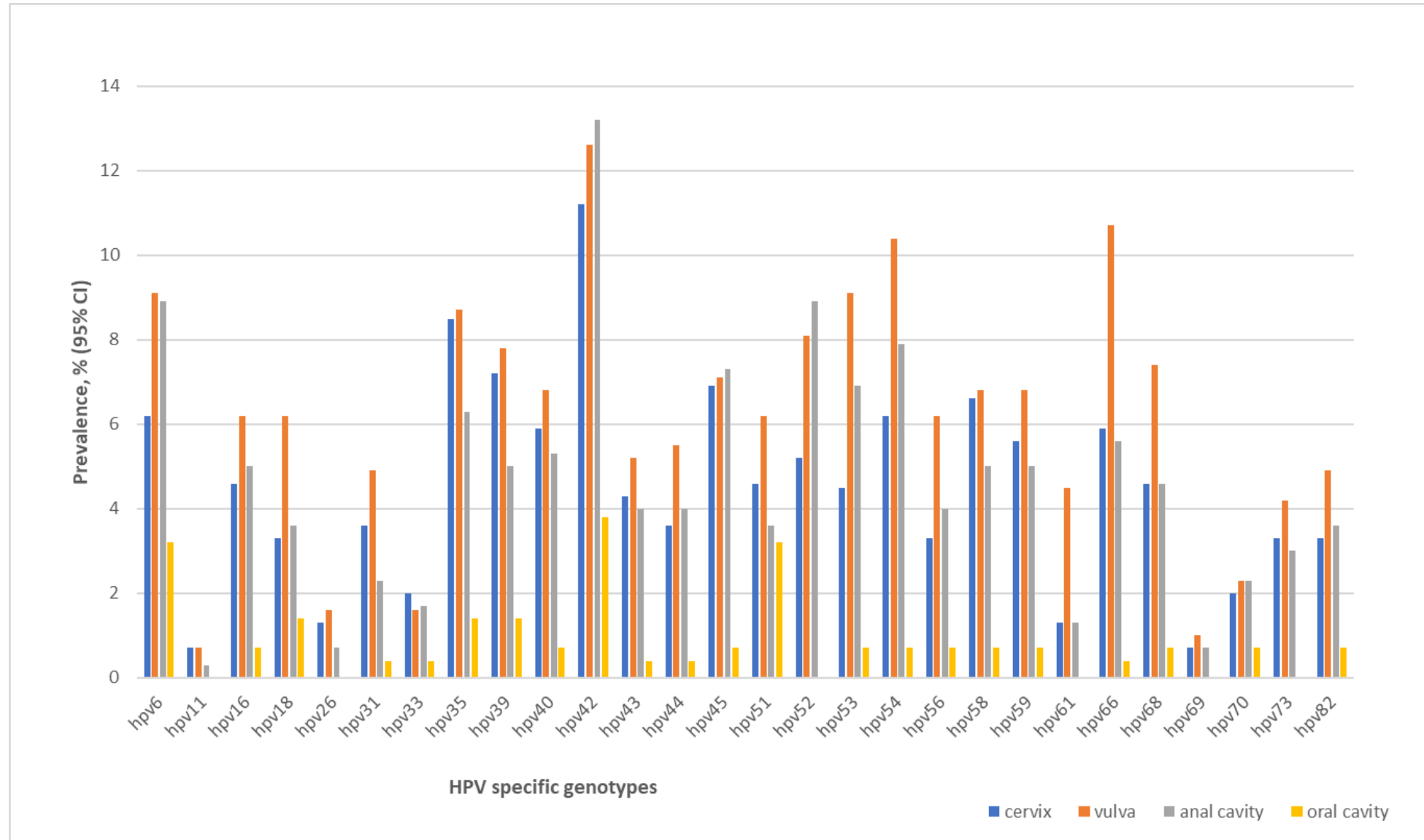
VARIABLE	GARDASIL [®]	CERVARIX [®]	GARDASIL [®] 9
Manufacturer	Merck and Co, Inc., Whitehouse Station, NJ, USA	GlaxoSmithKline Biologicals, Rixensart, Belgium	Merck and Co, Inc., Whitehouse Station, NJ, USA
Full name	Human papillomavirus Quadrivalent Vaccine, Recombinant	Human papillomavirus Bivalent Vaccine, Recombinant	Human Papillomavirus 9- valent Vaccine, Recombinant
HPV genotypes	6, 11, 16 and 18	16 and 18	6, 11, 16, 18, 31, 33, 45, 52 and 58
FDA approval date	2006	2008	2014
Recommended dose	2	2	3
Recommended dosing schedule	0, 6-12 months	0, 6-12 months	0, 1-2, 6 months
Prevention benefits	Mostly cervical cancer and warts	Mostly cervical cancer	Cervical, vulvar, vaginal, and anal cancers, and warts



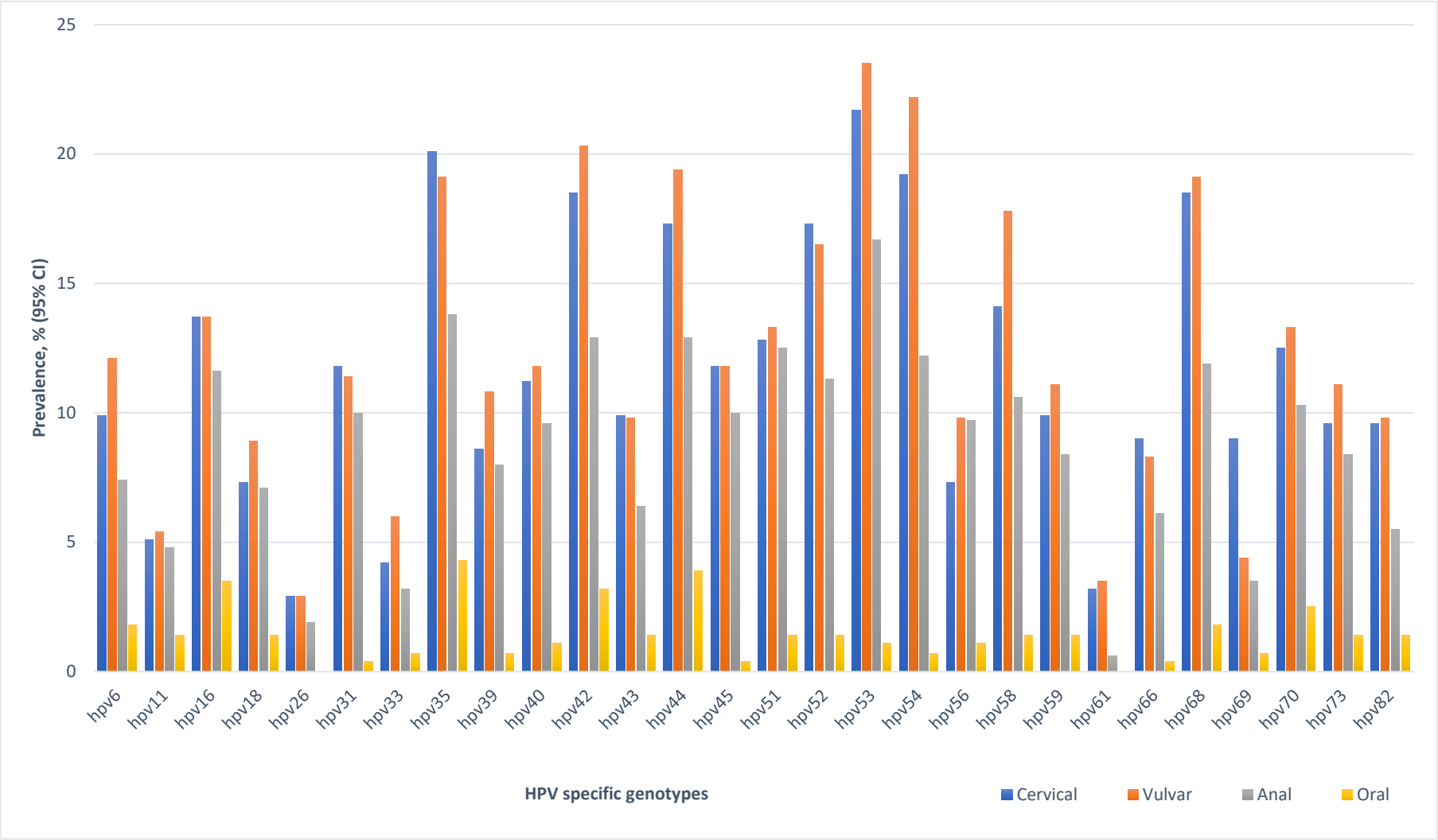
AVAILABLE SHINI SAMPLES

Study population	Cervix	Vulva	Anus	Penile	Oral	Serum	Urine
Women	310	310	310	-	310	310	310
Sex workers	316	316	316	-	316	316	316
Men	-	-	310	310	310	310	310

Prevalence of specific HPV genotypes according to the four anatomical sites in women



Prevalence of specific HPV genotypes according to the four anatomical sites among FSWs



Prevention of HPV associated Cancers

- **Primary prevention**
 - Behaviour Change Communication
 - Vaccine
- **Secondary prevention:** Screening methods
 - Cytology
 - Visual inspections
 - HPV testing
- **Tertiary prevention:** Detection & treatment of early cervical cancer

